MA1512 – Differential Equations for Engineering

A Book of Examples

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Contents

0 Before We Begin 5

Calculus of a Single Variable 5

The Hyperbolic Functions 8

The Complex Numbers 8

1 An Introduction to Ordinary Differential Equations 11

1.1 First Principles 11

1.2 Separation of Variables 14

1.3 The Geometry of Differential Equations 17

1.4 Population Dynamics 19

2 Linear Differential Equations 23

2.1 First-Order Linear Equations 23

2.2 Higher-Order Homogeneous Linear Equations 27

3 The Harmonic Oscillator 31

3.1 Non-Homogeneous Linear Differential Equations 31

3.2 Simple Harmonic Motion 37

4 The Laplace Transform 41

4.1 Basic Properties 41

4.2 Step Functions and the Unit Impulse 45

5 An Introduction to Partial Differential Equations 51

5.1 First Principles 51

5.2 The Heat Equation 55

Appendix 57

A Note on the Variation of Parameters 57

Table of Laplace Transforms 59

Properties of the Laplace Transform 60

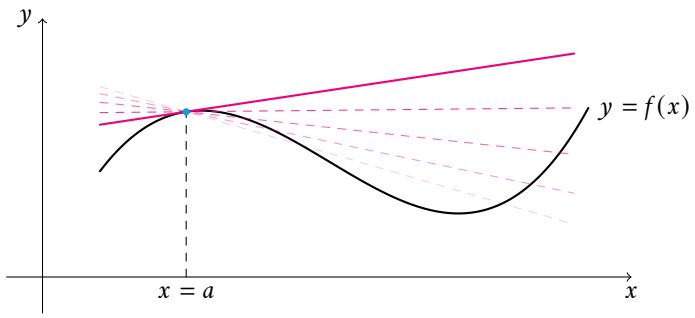
Index 63

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Before We Begin

Calculus of a Single Variable

The ideas and techniques of differential equations sit atop everything you have learned in mathematics thus far. At its heart, a differential equation is a statement about *derivatives.*

The derivative arises from the *tangent problem*: we wish to compute the slope of the tangent line to the graph of $y = f(x)$ at the point where $x = a$.



By drawing lines to points that get closer and closer to $x = a$, we can approximate the slope of the tangent line with the expression

$$\frac{f(x) - f(a)}{x - a} \quad \text{as} \quad x \to a.$$

This gives us the **derivative** of $y = f(x)$ at $x = a$, denoted $f'(a)$. The derivative admits a physical interpretation: it describes the *instantaneous rate of change* of $y = f(x)$ with respect to x when $x = a$. Formally defined, the **derivative** of a function f is given by

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}.$$

We use the following alternative notations for the derivative interchangeably:

- **prime notation:** $f'(x)$ or y', due to Joseph-Louis Lagrange;
- **Leibniz notation:** $\dfrac{dy}{dx}$ or $\dfrac{df}{dx}$ or $\dfrac{d}{dx}f$, due to Gottfried Wilhelm Leibniz;
- **dot notation:** $\dot{y}$, due to Isaac Newton, often used when the independent variable is time, i.e., for functions of the form $y(t)$.

We have the following results on derivatives:

- The derivative is linear:

$$(f+g)' = f' + g', \quad (cf)' = cf'.$$

- The derivative of a product is given by

$$(fg)' = f'g + fg'.$$

- The derivative of a quotient is given by

$$\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}.$$

The Chain Rule and Implicit Differentiation

The **Chain Rule** gives us a straightforward method of computing the derivative of a composite function. Suppose that g is differentiable at x and f is differentiable at $g(x)$. Then,

$$(f(g(x)))' = f'(g(x)) \cdot g'(x).$$

In Leibniz notation, writing $y = f(u)$ and $u = g(x)$, then

$$\frac{dy}{dx} = \frac{dy}{du}\frac{du}{dx}.$$

We will often encounter situations in which the variable y is not given explicitly as a function of x: instead, we have an *implicit* function of the form

$$F(x, y) = 0.$$

We can proceed to differentiate this function with respect to x, keeping the Chain Rule in mind whenever a term involves y. For instance,

$$\frac{d}{dx}y = \frac{dy}{dx}, \quad \frac{d}{dx}y^n = ny^{n-1}\frac{dy}{dx}, \quad \frac{d}{dx}e^y = e^y\frac{dy}{dx}, \quad \ldots.$$

More generally, we can differentiate a function of y with respect to x by first differentiating with respect to y, and then multiplying by dy/dx:

$$\frac{d}{dx}f(y) = \frac{df}{dy}\frac{dy}{dx}.$$

The Fundamental Theorem of Calculus

If $f(x) \geq 0$, the *area under the curve* $y = f(x)$ and above the x-axis from $x = a$ to $x = b$ is given by the **definite integral**

$$\int_a^b f(x)\, dx.$$

Surprisingly, the seemingly unrelated topics of differentiation (which investigates tangents) and integration (which finds areas) turn out to be closely related to one another: they are, in fact, inverse processes!

Theorem (The Fundamental Theorem of Calculus)**.** *Let $f(x)$ be a continuous function on an interval $[a, b]$. If*

$$g(x) = \int_a^x f(t)\, dt,$$

then

$$g'(x) = f(x).$$

Moreover, if G is any antiderivative of f, that is, $G' = f$, then

$$\int_a^b f(x)\, dx = G(b) - G(a).$$

We thus define the **indefinite integral** of a function $f(x)$ as the set of all antiderivatives of f: given $F'(x) = f(x)$,

$$\int f(x)\, dx = F(x) + c, \text{ where } c \in \mathbb{R}.$$

Tips for Evaluating Integrals

It is often useful to follow these steps, proceeding to the next one if you are left with an expression that is still difficult to integrate.

1. Simplify the integrand as much as possible: manipulate the expressions algebraically, or apply trigonometric identities. Sometimes, you can split the expression into a sum or difference of terms that are individually easier to integrate.
2. Look for an obvious substitution: find a function $u = g(x)$ whose differential $du = g'(x)\, dx$, up to a constant factor, also appears in the integral.
3. Classify the integrand's form, and use the appropriate substitution or decomposition method.
   - For rational functions, employ a partial fraction decomposition.
   - For products of polynomials and transcendental functions, try integration by parts.
   - For square roots, consider a trigonometric substitution.

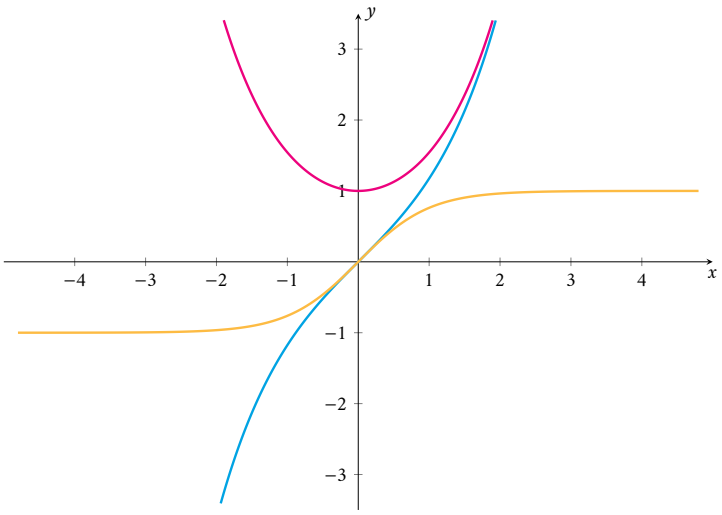
List of Common Integrals.

1. $\displaystyle\int u\, dv = uv - \int v\, du$
2. $\displaystyle\int x^n\, dx = \frac{x^{n+1}}{n+1} + c, n \neq -1$
3. $\displaystyle\int \frac{1}{x}\, dx = \ln|x| + c$
4. $\displaystyle\int e^x\, dx = e^x + c$
5. $\displaystyle\int \sin x\, dx = -\cos x + c$
6. $\displaystyle\int \cos x\, dx = \sin x + c$
7. $\displaystyle\int \sec^2 x\, dx = \tan x + c$
8. $\displaystyle\int \tan x\, dx = \ln|\sec x| + c$
9. $\displaystyle\int \frac{1}{\sqrt{a^2 - x^2}}\, dx = \sin^{-1}\frac{x}{a} + c$
10. $\displaystyle\int \frac{1}{a^2 + x^2}\, dx = \frac{1}{a}\tan^{-1}\frac{x}{a} + c$

The Hyperbolic Functions

The functions e^x and e^{-x}, and combinations of these two functions, come about in a variety of physical situations; as such, we define a set of functions—called the **hyperbolic functions**—which combine e^x and e^{-x} in some way. There are three main hyperbolic functions:

$$\sinh x = \frac{e^x - e^{-x}}{2}, \qquad \cosh x = \frac{e^x + e^{-x}}{2}, \qquad \tanh x = \frac{\sinh x}{\cosh x}.$$



As with the trigonometric functions, we can define the functions csch x, sech x, and coth x by taking the reciprocals of sinh x, cosh x, and tanh x respectively:

$$\operatorname{csch} x = \frac{1}{\sinh x}, \qquad \operatorname{sech} x = \frac{1}{\cosh x}, \qquad \coth x = \frac{\cosh x}{\sinh x}.$$

It is fairly straightforward to compute the derivatives of the hyperbolic functions from their definitions: for instance,

$$\frac{d}{dx}\sinh x = \frac{d}{dx}\left(\frac{e^x - e^{-x}}{2}\right) = \frac{e^x + e^{-x}}{2} = \cosh x,$$
$$\frac{d}{dx}\cosh x = \frac{d}{dx}\left(\frac{e^x + e^{-x}}{2}\right) = \frac{e^x - e^{-x}}{2} = \sinh x.$$

We leave you to compute the derivatives of the remaining hyperbolic functions and explore their various properties.

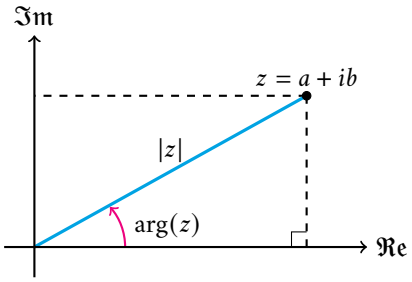
The Complex Numbers

The problems we encounter in this course are grounded on real physical systems—often, however, our computations are significantly simplified by working in the complex plane. We introduce the **imaginary unit** i, defined by the equation

$$i^2 = -1.$$

A **complex number** is a number of the form $z = a + ib$, where a and b are real numbers. Given a complex number $z = a + ib$ with $a, b \in \mathbb{R}$, we say that $a = \mathfrak{Re}\,(z)$ is the **real part** of z, and $b = \mathfrak{Im}\,(z)$ is the **imaginary part** of z. The **complex conjugate** of z is given by $z^* = a - ib$.

We can associate every complex number $a + ib$ to a point (a, b) on the *complex plane*: all real numbers with $b = 0$ lie on the (horizontal) *real axis*, and all purely imaginary numbers with $a = 0$ lie on the (vertical) *imaginary axis*.



Every complex number can thus be written in its **Cartesian form**

$$z = a + ib, \text{ with } a, b \in \mathbb{R}.$$

Given a complex number $z = a + ib$ with $a, b \in \mathbb{R}$,

- the **modulus** of z, denoted $|z|$, is the distance of z from the origin;
- the **argument** of z, denoted $\arg(z)$, is the angle made by $z = a + ib$ with respect to the positive real axis.

Note that $(r, \theta) = (|z|, \arg(z))$ are precisely the *polar coordinates* of the point with Cartesian coordinates $(x, y) = (a, b)$. Thus, every complex number $z = a + ib$ may also be written in **polar form** as

$$z = |z| (\cos\theta + i \sin\theta), \text{ where } \theta = \arg(z).$$

The prolific Swiss polymath Leonhard Euler (1707–1783) provides us with the following beautiful result.

Theorem (Euler's Formula)**.** *For any real number* θ*,*

$$e^{i\theta} = \cos\theta + i \sin\theta.$$

We may thus express every complex number in **exponential form**

$$z = |z| e^{i\theta}, \text{ where } \theta = \arg(z).$$

Using Euler's formula, we can rewrite complex exponentials in terms of real trigonometric functions:

$$e^{(\alpha+i\beta)x} = e^{\alpha x} e^{i\beta x} = (e^{\alpha x} \cos\beta x) + i (e^{\alpha x} \sin\beta x).$$

This identity plays a central role in our study of linear differential equations.

1
An Introduction to Ordinary Differential Equations

1.1 First Principles

A **differential equation** is an equation involving an (often unknown) function and its derivatives. We begin with a study of **ordinary differential equations**, which involve an independent variable (say, x), a function (say, $y\,(x)$), and one or more of its derivatives. The simplest way to describe a differential equation is in terms of its **order**: that is, the order of the highest derivative that appears in the equation.

Differential equations often arise when we try to express a physical process or some universal law in mathematical terms. The work of constructing a mathematical model often involves the following steps:

1. identifying the independent and dependent variables, and labelling them accordingly;
2. choosing the appropriate units of measurement for each variable; and
3. expressing the principle we wish to model as an equation in terms of our chosen variables, introducing physical constants if necessary.

When done correctly, all of the terms in our equation must have the same physical units.

Differential equations provide us with information about how a system changes *instantaneously*: 'short-term' information, often obtained through experimental evidence. Our challenge is to somehow use this information to determine what the system looks like in the 'long-term'. We do so by finding a **solution** to the differential equation: that is, we determine a function that satisfies the differential equation. When solving a differential equation, we may find a **general solution**: a family of infinitely many possible solutions, often involving arbitrary constants. With additional information (such as *initial conditions*), we can determine a **particular solution** that no longer involves any arbitrary constants.

Our practice of differential equations is thus concerned with the following process:

1. we first model a physical system by means of a differential equation,
2. we then solve the differential equation, and
3. we interpret our solutions in the physical context from which they arose.

Example 1. The decomposition of a radioactive element is proportional to the amount of substance present at any given time. Model this phenomenon by means of a differential equation, and find its solution.

Solution. Let $x(t)$ denote the amount of radioactive substance present at a given time t. Then,

$$\frac{dx}{dt} \propto x \quad \Longrightarrow \quad \frac{dx}{dt} = cx,$$

for some proportionality constant c. Since radioactive substances decrease in amount over time, this constant must be negative: we instead write

$$\frac{dx}{dt} = -kx, \text{ where } k > 0.$$

The constant k* is known as the **decay rate** of the element.

* Check that the units agree with one another: x is measured in units of mass and dx/dt in mass per unit time; thus, k must have units $[\text{time}]^{-1}$.

There is an 'obvious' solution: the constant *zero function* $x(t) = 0$. If $x(t) \neq 0$, we may divide throughout by x to yield

$$\frac{1}{x}\frac{dx}{dt} = -k.$$

We simplify the left-hand side by the Chain Rule: observe that

$$\frac{d}{dt}\ln|x| = \frac{1}{x}\cdot\frac{dx}{dt},$$

which is precisely the left-hand side of the above equation. Thus, we can integrate both sides of the differential equation with respect to t, yielding

$$\ln|x| = -kt + c, \text{ where } c \in \mathbb{R}.$$

Exponentiating both sides, we may equivalently write

$$x(t) = e^{-kt+c} \quad \Longrightarrow \quad x(t) = Ae^{-kt}, \text{ where } A \in \mathbb{R}.$$

Note that when $t = 0$, $x(0) = A$: thus, A gains a physical meaning—it represents the initial amount of substance present.

To determine k, additional information, such as the *half-life* of the radioactive substance, is required. If the half-life is known to be τ, then it takes τ amount of time for the substance to decrease by half: $x(\tau) = A/2$. Plugging this into our solution yields

$$\frac{A}{2} = Ae^{-k\tau} \quad \Longrightarrow \quad k = \frac{\ln 2}{\tau}.$$

Example 2. Experiments show that the rate at which an object's temperature changes is proportional to the difference between its temperature and that of its environment. Suppose that an object is placed in a room held at a constant ambient temperature of T_{env}.

1. Write down a differential equation that models the object's temperature $T(t)$ at any time t.

2. Show that a solution to this differential equation is given by the constant function
$$T(t) = T_{\text{env}}.$$

Solution. The rate at which $T(t)$ changes is proportional to the difference $T(t) - T_{\text{env}}$:
$$\frac{dT}{dt} \propto (T(t) - T_{\text{env}}) \quad \Longrightarrow \quad \frac{dT}{dt} = c\,(T(t) - T_{\text{env}}),$$
for some proportionality constant c. Note that

- when $T(t) > T_{\text{env}}$, that is, the object is at a temperature higher than that of its surroundings, we expect its temperature to decrease, and so $dT/dt < 0$; and

- when $T(t) < T_{\text{env}}$, that is, the object is colder than its surroundings, we expect its temperature to increase, and so $dT/dt > 0$.

In particular, dT/dt and $T(t) - T_{\text{env}}$ must have opposite signs, and so we expect c to be negative. We may thus write
$$\frac{dT}{dt} = -k\,(T(t) - T_{\text{env}}), \text{ where } k > 0.$$
To show that the function $T(t) = T_{\text{env}}$ is a solution to the differential equation, we just plug it into each side of the differential equation and show that equality holds. On the left-hand side, we have
$$\frac{dT}{dt} = \frac{d}{dt}T_{\text{env}} = 0,$$
and on the right-hand side, we have
$$-k\,(T(t) - T_{\text{env}}) = -k\,(T_{\text{env}} - T_{\text{env}}) = 0.$$
Indeed, the constant function $T(t) = T_{\text{env}}$ satisfies the differential equation. This makes sense—an object that is already at the same temperature as its surroundings will not cool down or warm up any further.

1.2 Separation of Variables

The simplest technique to solve a first-order differential equation is to exploit the Chain Rule: a first-order differential equation is **separable** if it can be written in the form

$$M(y)\frac{dy}{dx} = N(x).$$

To solve this, we directly integrate both sides of the equation with respect to x—computationally, it is often convenient to move the differentials and instead directly integrate both sides of the equation

$$M(y)\,dy = N(x)\,dx.$$

In some cases, a substitution is required in order to make a differential equation separable.

- If $y' = f(ax + by + c)$, we employ a *linear change of variable.* Let

$$u = ax + by + c \quad \Longrightarrow \quad u' = a + by'.$$

- If $y' = f(y/x)$, we let $y = xv$, and $y' = xv' + v$.

Example 3. Solve the differential equation

$$\frac{dy}{dx} = e^x\left(1 + y^2\right).$$

Solution. Since $1 + y^2 > 0$ for all $y \in \mathbb{R}$, we may divide both sides of the equation by $1 + y^2$. Separating the variables,

$$\int \frac{1}{1+y^2}\,dy = \int e^x\,dx \quad \Longrightarrow \quad \tan^{-1} y = e^x + c.$$

Example 4. Determine a particular solution to the initial value problem

$$\frac{dy}{dx} = \frac{xy^2}{x-1}, \quad y(2) = 1.$$

Solution. We first determine a general solution by performing a separation of variables. Note that we are given the condition $y(2) = 1$: that is, the solution $y(x)$ we wish to find is *not* the constant function $y(x) = 0$, and so we may proceed to divide both sides of the equation by y^2 :

$$\int \frac{1}{y^2}\,dy = \int \frac{x}{x-1}\,dx \quad \Longrightarrow \quad -\frac{1}{y} = \int \frac{x-1+1}{x-1}\,dx = \int 1 + \frac{1}{x-1}\,dx.$$

We thus find that the general solution is

$$-\frac{1}{y} = x + \ln|x-1| + c.$$

We can now determine a particular solution: to find the value of the arbitrary constant c, we substitute the initial condition $y(2) = 1$:

$$-1 = 2 + \ln|2-1| + c = 2 + 0 + c,$$

and hence $c = -3$, and the desired solution to the initial value problem is

$$-\frac{1}{y} = x + \ln|x-1| - 3.$$

Example 5. Find a general solution to the differential equation

$$\frac{dy}{dx} = (x+y)^2 .$$

Solution. We employ a *linear change of variable*: let

$$u = x + y \quad \Longrightarrow \quad \frac{du}{dx} = 1 + \frac{dy}{dx}.$$

Thus, our differential equation becomes

$$\frac{du}{dx} = 1 + u^2,$$

which is now separable in the variables x and u. Separating the variables, we have

$$\int \frac{1}{1+u^2}\, du = \int 1\, dx \quad \Longrightarrow \quad \tan^{-1} u = x + c.$$

Substituting back $u = x + y$, we arrive at the general solution

$$\tan^{-1}(x+y) = x + c \quad \Longrightarrow \quad x + y = \tan(x+c).$$

Example 6. Solve the differential equation

$$\frac{dy}{dx} = \frac{1-2y-4x}{1+y+2x}.$$

Solution. Observe that we may rewrite the differential equation as

$$\frac{dy}{dx} = \frac{1-2(y+2x)}{1+(y+2x)}.$$

We employ a *linear change of variable*: let

$$u = y + 2x \quad \Longrightarrow \quad \frac{du}{dx} = \frac{dy}{dx} + 2.$$

Thus, our differential equation becomes

$$\frac{du}{dx} = \frac{1-2u}{1+u} + 2 \quad \Longrightarrow \quad \frac{du}{dx} = \frac{3}{1+u}.$$

This is now a separable equation in x and u:

$$\int 1 + u\, du = \int 3\, dx \quad \Longrightarrow \quad u + \frac{u^2}{2} = 3x + c.$$

Since $u = y + 2x$, we thus have the general solution

$$y + 2x + \frac{(y+2x)^2}{2} = 3x + c.$$

Example 7. Find a general solution to the differential equation

$$\frac{dy}{dx} = e^{y/x} + \frac{y}{x} + 1.$$

Solution. We observe that the term y/x appears repeatedly throughout the differential equation. Consider the substitution $v = y/x$: that is,

$$y = xv \quad \Longrightarrow \quad \frac{dy}{dx} = v + x\frac{dv}{dx}.$$

Substituting this into the differential equation yields

$$v + x\frac{dv}{dx} = e^v + v + 1 \quad \Longrightarrow \quad x\frac{dv}{dx} = e^v + 1.$$

This is now a separable differential equation in the variables x and v:

$$\int \frac{1}{e^v + 1}\, dv = \int \frac{1}{x}\, dx.$$

We evaluate the integral on the left-hand side by substitution: let

$$u = e^v + 1 \quad \Longrightarrow \quad du = e^v\, dv = (u - 1)\, dv.$$

Then,

$$\int \frac{1}{e^v + 1}\, dv = \int \frac{1}{u} \cdot \frac{1}{u - 1}\, du = \int \frac{1}{u - 1} - \frac{1}{u}\, du = \ln|u - 1| - \ln|u| + c.$$

Note that $u = e^v + 1$; our differential equation thus has the solution

$$\ln|e^v| - \ln|e^v + 1| + c = \ln|x| \quad \Longrightarrow \quad \ln\left|\frac{e^v}{e^v + 1}\right| + c = \ln|x|\,.$$

Since $v = y/x$, a general solution to the differential equation is given by

$$\ln\left|\frac{e^{y/x}}{e^{y/x} + 1}\right| + c = \ln|x|\,.$$

We may leave our solution as is, or we may simplify further: exponentiating both sides and letting $A = e^c$, we have

$$A\frac{e^{y/x}}{e^{y/x} + 1} = x \quad \Longrightarrow \quad (A - x)\, e^{y/x} = x.$$

Example 8. Solve the differential equation

$$2xy\frac{dy}{dx} - y^2 + x^2 = 0.$$

Solution. We may rewrite the differential equation as

$$\frac{dy}{dx} = \frac{y^2 - x^2}{2xy} = \frac{1}{2}\left(\frac{y}{x} - \frac{x}{y}\right).$$

Consider the substitution $v = y/x$, or $y = xv$, such that $y' = v + xv'$. Substituting this into the differential equation yields

$$v + x\frac{dv}{dx} = \frac{1}{2}\left(v - \frac{1}{v}\right) \quad \Longrightarrow \quad x\frac{dv}{dx} = -\frac{1}{2}\left(v + \frac{1}{v}\right).$$

This is now a separable differential equation:

$$\int \frac{2v}{v^2 + 1}\, dv = \int -\frac{1}{x}\, dx \quad \Longrightarrow \quad \ln\left(v^2 + 1\right) = -\ln|x| + c.$$

Since $v = y/x$, we thus obtain the general solution

$$\ln\left|\frac{y^2}{x} + x\right| = c \quad \Longrightarrow \quad \frac{y^2}{x} + x = A.$$

1.3 The Geometry of Differential Equations

A lot of information can be gleaned from a differential equation without needing to solve it analytically. Consider a differential equation of the form

$$\frac{dy}{dx} = f(x, y).$$

From a geometric perspective, the above expression associates a derivative—that is, a slope—to every point (x, y) on the plane. Thus, a solution $y(x)$ to the differential equation is a function whose graph has the prescribed slope at each of the points it contains. That is, *the values of $f(x, y)$ yield the slopes of the tangents at all points on the graph of $y(x)$.*

We can thus gain an idea of what the graphs of the solutions $y(x)$ might look like by drawing a **direction field** (or a **slope field**): this is a graph representing how the solution $y(x)$ changes at various points. To sketch a direction field corresponding to a differential equation $y' = f(x, y)$, simply select various points (a, b) on the xy-plane; at each of these points, draw a short line segment whose slope is given by $f(a, b)$.

An **equilibrium solution** of a differential equation is a solution that is constant—these correspond to horizontal lines on a direction field, which occur precisely when $y' = 0$. An equilibrium solution $y(x) = \beta$ is said to be **stable** if solutions about this equilibrium approach β as $x \to \infty$. If solutions about the equilibrium move away from β as $x \to \infty$, the equilibrium is said to be **unstable**. If solutions lying on one side of the equilibrium tend to approach it, whereas solutions lying on the other side move away, the equilibrium is said to be **semistable**.

Example 9. Sketch a direction field for, and determine any equilibrium points of, the differential equation

$$y' = y.$$

Solution. If $y(x)$ is a solution to the differential equation, then the slope at every point (a, b) on the graph of $y(x)$ is simply b.

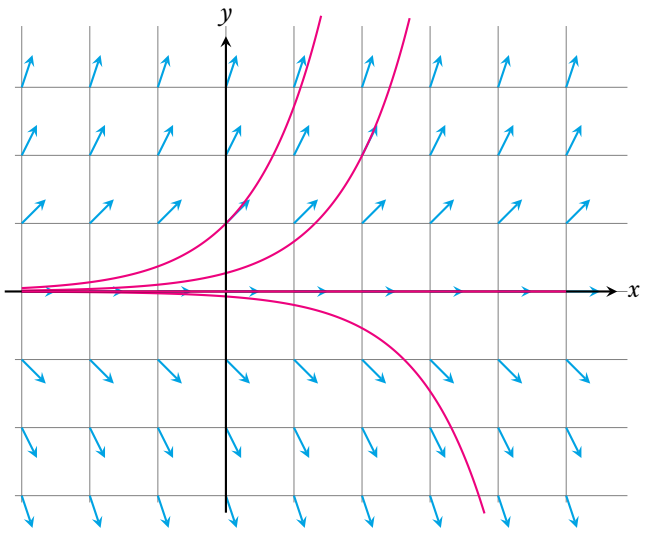


Figure 1: A slope field for $y' = y$. The graphs of particular solutions $y(x) = Ae^x$ for various values of A have been plotted in red.

The differential equation has an unstable equilibrium at $y = 0$. Moreover,

- when $y(0) > 0$, then $y' > 0$ and any solution $y(x)$ increases towards positive infinity;
- when $y(0) < 0$, then $y' < 0$ and any solution $y(x)$ decreases towards negative infinity.

Example 10. Consider the following differential equation:

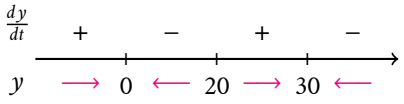
$$\frac{dy}{dt} = -10y(20 - y)\left(1 - \frac{1}{30}y\right).$$

1. Determine the equilibrium solutions of this differential equation, and state whether each equilibrium point is stable or unstable.
2. Determine the value of y as $t \to \infty$ when $y(0) = 15$. Does your answer change when $y(0) = 150$?

Solution. The equilibrium solutions occur when $dy/dt = 0$: we thus have $y = 0, y = 20$, and $y = 30$. Observe that the sign of dy/dt changes as we plug in different values of y :

$$\frac{dy}{dt} = \begin{cases} + & \text{when } y < 0 \text{ and } 20 < y < 30, \\ - & \text{when } 0 < y < 20 \text{ and } y > 30. \end{cases}$$

We can sketch out a sign diagram for dy/dt.



Thus, if $y(0) = 15$, we find that $y' < 0$ and y decreases towards the equilibrium at 0 as $t \to \infty$. When $y(0) = 150$, we find that $y' < 0$ and y decreases towards the equilibrium at 30 as t increases.

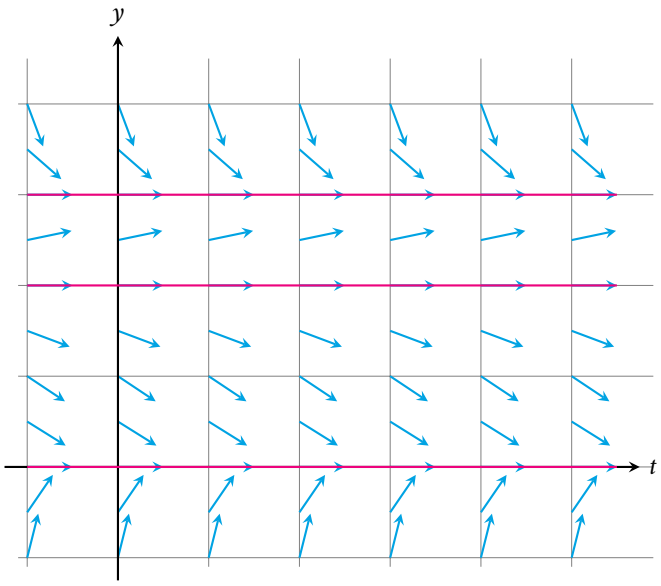


Figure 2: A sketch of the slope field for the given differential equation. The equilibrium solutions have been graphed in red.

In particular, we have stable equilibria at $y = 0$ and $y = 30$, and an unstable equilibrium at $y = 20$.

1.4 Population Dynamics

We can talk about rates of change and equilibria in more concrete terms by looking at some models of population.

Let $y(t)$ denote the population of a certain species at any given time t. The simplest hypothesis concerning the variation of population, attributed to the 18th century English economist Thomas Malthus, is that the rate of change of y is proportional to the present value of y: we have the **exponential (Malthus) model**

$$\frac{dy}{dt} = ky \quad \Longrightarrow \quad y(t) = y_0 e^{kt}, \text{ where } y_0 = y(0).$$

This model assumes that, in the absence of limiting factors, a population would grow exponentially with *growth rate* k.

To account for the possibility of the growth rate depending on the population, we replace the constant k in the differential equation with a function of y instead:

$$\frac{dy}{dt} = f(y)\, y.$$

In a world with *finite* resources, we might hypothesize that very large populations will lead to starvation and disease—we can thus come up with a function such that $f(y) \approx k > 0$ when the population is small, and decreases as y grows larger, eventually becoming $f(y) < 0$ when y is sufficiently large. After the 19th century Belgian mathematician Pierre Verhulst, we have the **logistic (Verhulst) model**

$$\frac{dy}{dt} = (k - ay)\, y = k\left(1 - \frac{y}{\beta}\right) y \text{ for some } a > 0.$$

This model assumes that a population grows logistically, such that given any initial population, the solution tends towards the stable equilibrium

$$\lim_{t\to\infty} y(t) = \beta.$$

This constant β, often denoted y_∞, is called the **carrying capacity** of the species.

Example 11. Solve Verhulst's differential equation given a known initial population $y(0) = y_0$.

Solution. We perform a separation of variables:

$$\int \frac{1}{y\,(1 - y/y_\infty)}\, dy = \int k\, dt.$$

To evaluate the integral on the left-hand side, we perform a partial fraction decomposition

$$\int \frac{1}{y} + \frac{1}{y_\infty - y}\, dy = \int k\, dt \quad \Longrightarrow \quad \ln|y| - \ln|y_\infty - y| = kt + c.$$

Combining terms and exponentiating yields

$$\frac{y_\infty}{y} - 1 = Ae^{-kt} \quad \Longrightarrow \quad y(t) = \frac{y_\infty}{1 + Ae^{-kt}}.$$

When $t = 0$, we have $y = y_0$, so $A = y_\infty/y_0 - 1$. Thus, we find that

$$y(t) = \frac{y_\infty}{1 + (y_\infty/y_0 - 1)\, e^{-kt}}.$$

Example 12. A certain population behaves according to the Verhulst model. Suppose that a constant number E of this species is captured per unit time. Model this situation as a differential equation. Determine any equilibrium solutions, and hence, find the largest possible number E of species that can be captured such that the population does not go extinct.

Solution. We directly modify the Verhulst model to account for the harvesting of species:

$$\frac{dy}{dt} = k\left(1 - \frac{y}{y_\infty}\right)y - E = -\frac{k}{y_\infty}y^2 + ky - E.$$

The equilibria of this differential equation can be computed with the quadratic formula:

$$y = \frac{-k \pm \sqrt{k^2 - 4kE/y_\infty}}{-2k/y_\infty} = \frac{k \mp \sqrt{k^2 - 4kE/y_\infty}}{2k/y_\infty}.$$

If $k^2 - 4kE/y_\infty < 0$, then the differential equation has no equilibrium points. In particular, dy/dt will always be negative: any solution to the differential equation will be a decreasing function, and the population is bound to disappear. We thus require that

$$k^2 - 4kE/y_\infty \geq 0 \quad \Longrightarrow \quad E \leq ky_\infty/4,$$

and we can capture at most $E = ky_\infty/4$ of the species per unit time without causing the population to go extinct. We now have a few different possibilities for the equilibria of the differential equation:

Case 1. If $E = ky_\infty/4$, there is one equilibrium point at $\beta = y_\infty/2$. This is a *semistable* equilibrium: if $y_0 > \beta$, then the population decreases towards the equilibrium; however, if $y_0 < \beta$, the population decreases rapidly until extinction.

Case 2. If $E < ky_\infty/4$, then the differential equation has two equilibrium points:

$$\beta_1 = \frac{k - \sqrt{k^2 - 4kE/y_\infty}}{2k/y_\infty}, \quad \beta_2 = \frac{k + \sqrt{k^2 - 4kE/y_\infty}}{2k/y_\infty}.$$

The equilibrium point at $y = \beta_2$ is *stable*, since small fluctuations about this point will allow the population to return to this equilibrium. On the other hand, $y = \beta_1$ is an *unstable* equilibrium since a fluctuation about this point can potentially lead the population to extinction.

We may visualize the populations' trajectories by roughly sketching out some slope fields of the differential equation for various values of E, based on the signs (i.e., positivity or negativity) of dy/dt.

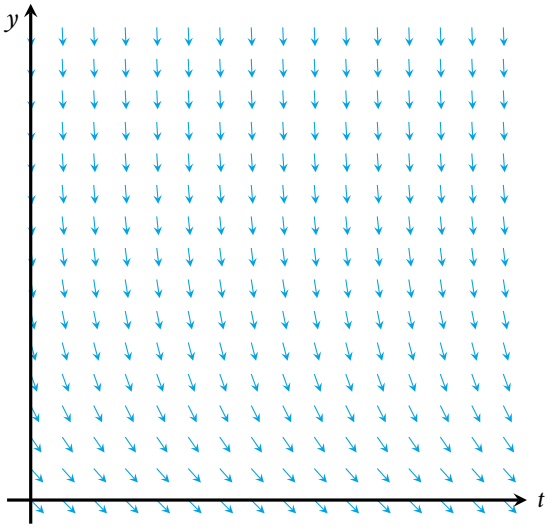
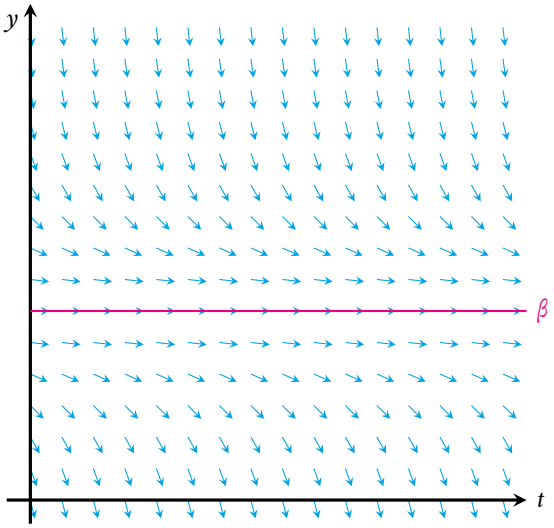
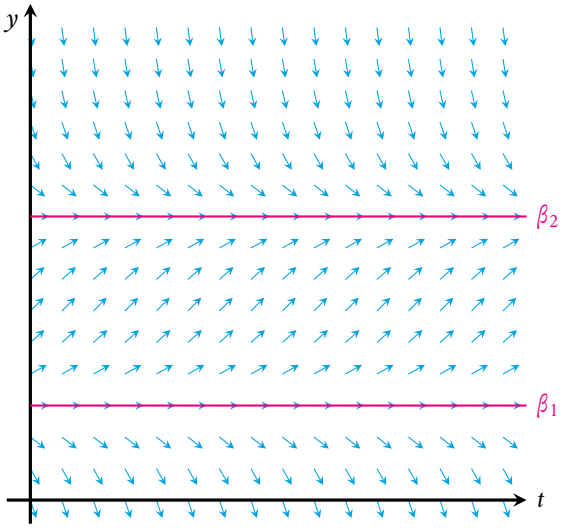


Figure 3: Sketches of slope fields of the modified Verhulst model for various values of E.

When $E > ky_\infty/4$, the differential equation has no equilibria, and the population is always bound for extinction.



When $E = ky_\infty/4$, there is one semistable equilibrium point, and slight fluctuations in the population may lead to extinction.



When $E < ky_\infty/4$, there are two equilibria. At the stable equilibrium β_2, slight fluctuations in the population will allow the species to recover.

2
Linear Differential Equations

2.1 First-Order Linear Equations

We now investigate a broad class of differential equations that admit systematic methods for finding their solutions. A **linear differential equation** is an equation of the form

$$a_n(x)\,y^{(n)} + a_{n-1}(x)\,y^{(n-1)} + \cdots + a_1(x)\,y' + a_0(x)\,y = f(x).$$

Here, the (unknown) functions $y, y', y'', \ldots$ appear *linearly*.

A **first-order linear** differential equation is an equation of the form

$$a(x)\,y' + b(x)\,y = c(x), \text{ with } a(x) \neq 0.$$

We can always solve this differential equation by the method of **integrating factors**.

1. Rewrite the entire equation in standard linear form:

$$y' + p(x)\,y = q(x).$$

2. Calculate the integrating factor* $u = e^{\int p(x)\,dx}$, and multiply it to both sides of the equation:

$$u(y' + py) = uq \quad \Longrightarrow \quad (uy)' = uq.$$

3. Integrate both sides of the equation.

* This has been specially chosen to ensure that the left-hand side of the equation simplifies by the Product Rule.

In some cases, we may encounter equations that look almost linear: a **Bernoulli differential equation**, named after the Swiss mathematician Jacob Bernoulli (1654–1705), is an equation which can be expressed in the form

$$y' + p(x)\,y = q(x)\,y^n \quad \equiv \quad y^{-n}y' + y^{1-n}p(x) = q(x).$$

To solve this, we consider the substitution $v = y^{1-n}$, so $v' = (1-n)\,y^{-n}y'$. Then, the Bernoulli equation is reduced to a first-order linear differential equation

$$v' + v(1-n)\,p(x) = q(x)(1-n),$$

which can now be solved using the method of integrating factors.

Example 13. Find a general solution to the differential equation

$$x y' - y = x^3.$$

Solution. Observe that the differential equation is linear, but not separable—we thus solve it by using an integrating factor. First, we rewrite it in standard linear form: dividing throughout by x, we have

$$y' - \frac{1}{x} y = x^2.$$

We calculate the integrating factor: observe that $p(x) = -1/x$, and so our integrating factor is

$$\int p(x)\, dx = -\ln x \quad \Longrightarrow \quad u = e^{-\ln x} = \frac{1}{x}.$$

Multiplying both sides of the equation by the integrating factor yields

$$\frac{1}{x}\left(y' - \frac{1}{x} y\right) = \frac{1}{x}\left(x^2\right) \quad \Longrightarrow \quad \left(\frac{1}{x} y\right)' = x.$$

Integrating both sides with respect to x then yields the general solution

$$\frac{1}{x} y = \frac{x^2}{2} + c.$$

Example 14. Find a particular solution to the initial value problem

$$x y' + 3y = \frac{\sin x}{x^2}, \quad y(\pi/2) = 1.$$

Solution. We have a linear differential equation: dividing throughout by x, we obtain its standard form

$$y' + \frac{3}{x} y = \frac{\sin x}{x^3}.$$

We calculate the integrating factor, with $p(x) = 3/x$:

$$\int p(x)\, dx = 3 \ln x \quad \Longrightarrow \quad u = e^{3 \ln x} = x^3.$$

Multiplying both sides of the equation by the integrating factor, we have

$$x^3\left(y' + \frac{3}{x} y\right) = x^3\left(\frac{\sin x}{x^3}\right) \quad \Longrightarrow \quad \left(x^3 y\right)' = \sin x.$$

Integrating yields the general solution

$$x^3 y = -\cos x + c.$$

We now determine the particular solution: plugging in $x = \pi/2$ and $y = 1$,

$$(\pi/2)^3 \cdot 1 = -\cos(\pi/2) + c \quad \Longrightarrow \quad c = \frac{\pi^3}{8}.$$

Hence, the solution to the initial value problem is given by

$$x^3 y = -\cos x + \frac{\pi^3}{8}.$$

Example 15. Find a general solution to the differential equation

$$xy' - y = -xy^2.$$

Solution. Observe that the differential equation is Bernoulli, since it can be written in the form

$$y' - \frac{1}{x}y = -y^2.$$

Dividing throughout by y^2, we have

$$y^{-2}y' - \frac{1}{x}y^{-1} = -1.$$

Thus, we let $v = y^{-1}$, and $v' = -y^{-2}\,y'$. Hence,

$$-v' - \frac{1}{x}v = -1 \quad \Longrightarrow \quad v' + \frac{1}{x}v = 1.$$

This is a linear equation with integrating factor $e^{\int 1/x\,dx} = x$. Hence,

$$x \cdot \left(v' + \frac{1}{x}v\right) = x \quad \Longrightarrow \quad (xv)' = x \quad \Longrightarrow \quad xv = \frac{1}{2}x^2 + c.$$

Substituting $v = y^{-1}$ back, we end up with

$$\frac{x}{y} = \frac{1}{2}x^2 + c.$$

Example 16. Solve the following differential equation:

$$y' + 2xy = -8x^3y^3.$$

Solution. This is a Bernoulli equation: dividing throughout by y^3 yields

$$y^{-3}y' + 2xy^{-2} = -8x^3.$$

Let $v = y^{-2}$, and $v' = -2y^{-3}y'$. Substituting this into the equation yields

$$-\frac{1}{2}v' + 2xv = -8x^3 \quad \Longrightarrow \quad v' - 4xv = 16x^3.$$

This is a linear equation with integrating factor $e^{\int -4x\,dx} = e^{-2x^2}$, and so

$$e^{-2x^2}\,(v' - 4xv) = e^{-2x^2}\,\left(16x^3\right) \quad \Longrightarrow \quad \left(e^{-2x^2}v\right)' = 16x^3e^{-2x^2}.$$

We integrate the right-hand side by substitution: let $u = x^2$, and $du = 2x\,dx$. Then,

$$\int 16x^3e^{-2x^2}\,dx = \int 8ue^{-2u}\,du = -2e^{-2u}\,(2u + 1) + c.$$

Since $u = x^2$, integrating both sides of our differential equation yields

$$e^{-2x^2}v = -2e^{-2x^2}\left(2x^2 + 1\right) + c.$$

Substituting back $v = y^{-2}$ and multiplying throughout by e^{2x^2}, we have the general solution

$$y^{-2} = -4x^2 - 2 + ce^{2x^2}.$$

Example 17. An object of mass m, initially at rest, is dropped from a certain height. As it falls towards the earth, it experiences a downward gravitational force, as well as an upward drag force. It is known that this force due to air resistance is directly proportional to the object's speed. Use Newton's Second Law to set up a differential equation modeling this system. Hence, determine the downward displacement x travelled by the object as a function of time t.

Solution. By Newton's Second Law, the sum of forces acting on an object is equal to the product of the object's mass and acceleration, $F = ma$, where $a(t)$ denotes the object's acceleration at time t. In this case, we have two forces acting on the object: we have

- a force due to gravity, $F_{\text{gravity}} = mg$, where $g = 9.80665\ \text{m s}^{-2}$ is the constant acceleration due to gravity, and
- a drag force in the opposite direction: $F_{\text{drag}} = -kv$, where $v(t)$ denotes the velocity of the object at time t, and k is some constant.

Since acceleration is the first time derivative of velocity, we thus have the differential equation

$$mv' = mg - kv.$$

This is a linear differential equation: writing it in standard form, we have

$$v' + \frac{k}{m}v = g.$$

We have the integrating factor $e^{\int k/m\,dt} = e^{kt/m}$. Multiplying this to both sides of the equation yields

$$e^{kt/m}\left(v' + \frac{k}{m}v\right) = e^{kt/m}g \quad \Longrightarrow \quad \left(e^{kt/m}v\right)' = e^{kt/m}g.$$

Integrating both sides of the equation with respect to t yields

$$e^{kt/m}v = \frac{mg}{k}e^{kt/m} + c.$$

Since the object starts at rest, we know that $v(0) = 0$. Thus, $c = -mg/k$, and we have the solution to the differential equation

$$v(t) = \frac{mg}{k} - \frac{mg}{k}e^{-kt/m} = \frac{mg}{k}\left(1 - e^{-kt/m}\right).$$

To determine the distance $x(t)$ travelled by the object, we can simply integrate our expression for the velocity:

$$x(t) = \int \frac{mg}{k}\left(1 - e^{-kt/m}\right)dt = \frac{mg}{k}\left(t + \frac{m}{k}e^{-kt/m}\right) + c.$$

Setting the starting point as the origin, we have $x(0) = 0$, and so $c = -m^2g/k^2$, and the displacement of the object is given by

$$x(t) = \frac{mg}{k}\left(t + \frac{m}{k}\left(e^{-kt/m} - 1\right)\right).$$

Note that the differential equation can also be solved by a separation of variables—try it out!

2.2 Higher-Order Homogeneous Linear Equations

Let us now consider a higher-order linear differential equation: suppose that our equation has *constant coefficients*, such that it is of the form

$$a_n y^{(n)} + a_{n-1} y^{(n-1)} + \cdots + a_1 y' + a_0 y = f(x), \text{ where } a_i \in \mathbb{R}.$$

When $f(x) = 0$, the equation is said to be **homogeneous**; otherwise, the differential equation is **non-homogeneous**.

In the case that the equation is homogeneous, we have the following property that arises from the basic properties of differentiation.

Theorem (Superposition Principle)**.** *Let $y_1(x)$ and $y_2(x)$ be solutions to a homogeneous linear differential equation*

$$a_n y^{(n)} + a_{n-1} y^{(n-1)} + \cdots + a_1 y' + a_0 y = 0.$$

Then, a solution to the differential equation is also given by

$$y(x) = c_1 y_1(x) + c_2 y_2(x), \text{ for all } c_1, c_2 \in \mathbb{R}.$$

To determine solutions to a homogeneous differential equation, we make a guess: we try

$$y(x) = e^{\lambda x}.$$

Plugging this into the differential equation yields its **characteristic equation**:

$$a_n \lambda^n + a_{n-1} \lambda^{n-1} + \cdots + a_1 \lambda + a_0 = 0.$$

- If $\lambda \in \mathbb{R}$ is a real, distinct root, then a solution is given by

$$e^{\lambda x}.$$

- If $\lambda \in \mathbb{R}$ is a repeated root with multiplicity r, then solutions are obtained by modifying our trial solution by a factor of x:

$$e^{\lambda x},\ xe^{\lambda x},\ x^2 e^{\lambda x},\ \ldots,\ x^{r-1} e^{\lambda x}.$$

- If $\lambda, \lambda^* \in \mathbb{C}$ are conjugate roots $\alpha \pm i\beta$, then Euler's formula yields

$$e^{(\alpha \pm i\beta)x} = e^{\alpha x} \cos \beta x \pm i e^{\alpha x} \sin \beta x.$$

  Extracting the real and imaginary parts yields the solutions

$$e^{\alpha x} \cos \beta x,\ e^{\alpha x} \sin \beta x.$$

  If $\lambda, \lambda^* \in \mathbb{C}$ are *repeated* roots, then we can extend the result on real repeated roots by modifying our solutions with a factor of x:

$$e^{\alpha x} \cos \beta x,\ xe^{\alpha x} \cos \beta x,\ x^2 e^{\alpha x} \cos \beta x,\ \ldots,$$
$$e^{\alpha x} \sin \beta x,\ xe^{\alpha x} \sin \beta x,\ x^2 e^{\alpha x} \sin \beta x,\ \ldots.$$

Example 18. Find the general solution to

$$4y'' + 12y' + 9y = 0.$$

Solution. The characteristic equation is

$$4\lambda^2 + 12\lambda + 9 = (2\lambda + 3)^2 = 0,$$

with a repeated root $\lambda = -3/2$. Thus, solutions are given by $e^{-\frac{3}{2}x}$ and $xe^{-\frac{3}{2}x}$, and the general solution is given by

$$y = c_1 e^{-\frac{3}{2}x} + c_2 x e^{-\frac{3}{2}x}.$$

Example 19. Find the general solution to

$$y'' - 6y' + 13y = 0.$$

Solution. The characteristic equation is

$$\lambda^2 - 6\lambda + 13 = 0.$$

By the quadratic formula, we find complex conjugate roots:

$$\lambda = \frac{6 \pm \sqrt{36 - 52}}{2} = 3 \pm 2i.$$

Thus, solutions are given by $e^{3x}\cos 2x$ and $e^{3x}\sin 2x$, and we may form the general solution

$$y = c_1 e^{3x}\cos 2x + c_2 e^{3x}\sin 2x.$$

Example 20. Find the particular solution to the following initial value problem:

$$y'' - y = 0, \quad y(0) = 5, \quad y'(0) = 3.$$

Solution. The characteristic equation is

$$\lambda^2 - 1 = (\lambda + 1)(\lambda - 1) = 0,$$

which has distinct roots $\lambda = -1, 1$. We thus have the general solution

$$y(x) = c_1 e^{-x} + c_2 e^{x}.$$

To determine the values of c_1 and c_2, we plug in the initial conditions. Note that

$$y'(x) = -c_1 e^{-x} + c_2 e^{x}.$$

Thus, when $x = 0$,

$$y(0) = c_1 + c_2 = 5, \quad y'(0) = -c_1 + c_2 = 3.$$

We can now solve for the values of $c_1$and c_2:

$$\left\{\begin{array}{rcrcl} c_1 & + & c_2 & = & 5 \\ -c_1 & + & c_2 & = & 3 \end{array}\right. \implies \left\{\begin{array}{rcrcl} c_1 & + & c_2 & = & 5 \\ & & 2c_2 & = & 8 \end{array}\right. .$$

Hence, $c_2 = 4$ and $c_1 = 1$, and the particular solution is thus

$$y(x) = e^{-x} + 4e^{x}.$$

Example 21. Find the particular solution to the following initial value problem:

$$y'' - 4y' + 20y = 0, \quad y(\pi/2) = 0, \quad y'(\pi/2) = 1.$$

Solution. The characteristic equation is

$$\lambda^2 - 4\lambda + 20 = 0.$$

By the quadratic formula, its roots are given by

$$\lambda = \frac{4 \pm \sqrt{16 - 80}}{2} = 2 \pm 4i.$$

Solutions are thus given by $e^{2x}\cos 4x$ and $e^{2x}\sin 4x$, and we have the general solution

$$y(x) = c_1 e^{2x}\cos 4x + c_2 e^{2x}\sin 4x.$$

To determine the values of c_1 and c_2, we plug in the given conditions. We have

$$y'(x) = c_1\left(2e^{2x}\cos 4x - 4e^{2x}\sin 4x\right) + c_2\left(2e^{2x}\sin 4x + 4e^{2x}\cos 4x\right).$$

Thus, when $x = \pi/2$,

$$y(\pi/2) = c_1 e^{\pi} = 0, \quad y'(\pi/2) = 2e^{\pi}c_1 + 4e^{\pi}c_2 = 1.$$

Thus, $c_1 = 0$ and $c_2 = 1/4e^{\pi}$, and we have the particular solution

$$y(x) = \frac{1}{4}e^{2x-\pi}\sin 4x.$$

Example 22. Determine the general solution of the following *fourth*-order differential equation

$$y^{(4)} - 5y'' - 36y = 0.$$

Solution. The characteristic equation is

$$\lambda^4 - 5\lambda^2 - 36 = \left(\lambda^2 - 9\right)\left(\lambda^2 + 4\right) = 0.$$

It has four roots:

- two real roots $\lambda = -3, 3$, from which we have solutions e^{-3x} and e^{3x};
- complex conjugate roots $\lambda = \pm 2i$, yielding $\cos 2x$ and $\sin 2x$.

We can thus construct the general solution:

$$y = c_1 e^{-3x} + c_2 e^{3x} + c_3\cos 2x + c_4\sin 2x.$$

3
The Harmonic Oscillator

3.1 Non-Homogeneous Linear Differential Equations

We now consider the problem of solving *non-homogeneous* linear differential equations.

Theorem. *Suppose that y_p is a particular solution to the differential equation*

$$a_n(x)\,y^{(n)} + a_{n-1}(x)\,y^{(n-1)} + \cdots + a_1(x)\,y' + a_0(x)\,y = f(x).$$

Then, a general solution to the differential equation can be written in the form

$$y(x) = y_h(x) + y_p(x),$$

where $y_h(x)$ is the general solution to the complementary homogeneous equation

$$a_n(x)\,y^{(n)} + a_{n-1}(x)\,y^{(n-1)} + \cdots + a_1(x)\,y' + a_0(x)\,y = 0.$$

Consider, for simplicity, the second-order linear equation with constant coefficients

$$y'' + py' + qy = f(x), \quad \text{where} \quad f(x) \neq 0.$$

When $f(x)$ involves simple functions, we can attempt to 'guess' a particular solution $y_p(x)$ by the **method of undetermined coefficients**. For example,

- if $f(x)$ is a polynomial of degree n, we let y_p be an arbitrary nth degree polynomial with undetermined coefficients;
- if $f(x)$ contains an exponential e^{kx}, then we try $y_p = Ae^{kx}$, where $A \in \mathbb{R}$ is an undetermined coefficient;
- if $f(x)$ contains either $\cos kx = \mathfrak{Re}(e^{ikx})$ or $\sin kx = \mathfrak{Im}(e^{ikx})$, we can instead solve for the real or imaginary solutions, respectively, of the complex equation
$$z'' + pz' + qz = e^{ikx}$$
by trying $z_p = Ae^{ikx}$, where $A \in \mathbb{C}$ is an undetermined coefficient.

If any term of the trial solution is already a solution of the complementary equation, we modify our trial solution by multiplying by a factor of x.

Alternatively, performing a **variation of parameters** provides a more analytic approach to determining a particular solution $y_p(x)$: suppose that the general solution to the complementary equation has the form

$$y_h(x) = c_1 y_1(x) + c_2 y_2(x).$$

We vary the parameters c_1 and c_2 to construct a particular solution*

$$y_p(x) = u(x)\, y_1(x) + v(x)\, y_2(x),$$

where the functions $u(x)$ and $v(x)$ are given by

$$u(x) = -\int \frac{y_2 f}{y_1 y_2' - y_1' y_2}\, dx, \qquad v(x) = \int \frac{y_1 f}{y_1 y_2' - y_1' y_2}\, dx.$$

* See "A Note on the Variation of Parameters" in the Appendix for a proof.

Example 23. Find the general solution to

$$y'' + 4y = 2e^{2x}.$$

Solution. The characteristic equation of the complementary equation is $\lambda^2 + 4 = 0$, with complex roots $\lambda = \pm 2i$, and so

$$y_h = c_1 \cos 2x + c_2 \sin 2x.$$

We now find a particular solution: since $f(x) = 2e^{2x}$, we might guess

$$y_p = Ae^{2x}.$$

Then, $y_p' = 2Ae^{2x}$ and $y_p'' = 4Ae^{2x}$. Substituting these into the equation,

$$4Ae^{2x} + 4\left(Ae^{2x}\right) = 2e^{2x} \quad \Longrightarrow \quad 8A = 2 \quad \Longrightarrow \quad A = \frac{1}{4}.$$

Hence, we have the particular solution $y_p = e^{2x}/4$, and we obtain the general solution

$$y = c_1 \cos 2x + c_2 \sin 2x + \frac{e^{2x}}{4}.$$

Example 24. Find the general solution to

$$y'' + y' - 2y = 4x^2.$$

Solution. We begin by solving the complementary equation $y'' + y' - 2y = 0$. The characteristic equation is

$$\lambda^2 + \lambda - 2 = (\lambda - 1)(\lambda + 2) = 0 \quad \Longrightarrow \quad \lambda = 1, -2.$$

Hence, we have the solution of the complementary equation

$$y_h = c_1 e^{x} + c_2 e^{-2x}.$$

We now find a particular solution: observe that $f(x) = 4x^2$ is a polynomial of degree 2, so we might guess a particular solution to be a polynomial of degree 2, as well: such a polynomial would take on the form

$$y_p = Ax^2 + Bx + C.$$

We now determine the coefficients A, B, and C. Observe that $y_p' = 2Ax + B$, and $y_p'' = 2A$. Substituting this into the differential equation,

$$2A + 2Ax + B - 2\left(Ax^2 + Bx + C\right) = 4x^2$$
$$-2Ax^2 + (2A - 2B)\, x + (2A + B - 2C) = 4x^2.$$

Comparing coefficients, we find that

$$-2A = 4, \quad 2A - 2B = 0, \quad 2A + B - 2C = 0.$$

We find that $A = -2$, $B = -2$, and $C = -3$, so $y_p = -2x^2 - 2x - 3$. Hence, the general solution is

$$y = y_h + y_p = c_1 e^{x} + c_2 e^{-2x} - 2x^2 - 2x - 3.$$

Example 25. Find a general solution to

$$y'' + y' - 2y = \sin x.$$

Solution. From the previous example, we have

$$y_h = c_1 e^x + c_2 e^{-2x}.$$

We now find a particular solution: note that both cosine and sine terms can be differentiated to yield $\sin x$. In order to minimize dealing with sines and cosines, we recall Euler's formula:

$$e^{ix} = \cos x + i \sin x.$$

In particular, $\sin x = \mathfrak{Im}\left(e^{ix}\right)$. Hence, we can equivalently solve for the particular solution of the complex differential equation

$$z'' + z' - 2z = e^{ix},$$

noting that we only want the imaginary part of the solution.

Since $f(x) = e^{ix}$, we guess

$$z_p = Ae^{ix}.$$

Since i is simply a constant, the derivatives are given by

$$z_p' = Aie^{ix}, \quad z_p'' = -Ae^{ix}.$$

Plugging this into the differential equation yields

$$\underbrace{-Ae^{ix}}_{z''} + \underbrace{Aie^{ix}}_{z'} - \underbrace{2Ae^{ix}}_{2z} = e^{ix} \quad \Longrightarrow \quad A(-3+i) = 1.$$

Thus, we find that

$$A = \frac{1}{-3+i} = \frac{1}{-3+i} \cdot \frac{-3-i}{-3-i} = -\frac{3}{10} - \frac{1}{10}i.$$

We thus have the complex solution

$$\begin{aligned} z_p &= \left(-\frac{3}{10} - \frac{1}{10}i\right)(\cos x + i \sin x) \\ &= -\frac{3}{10}\cos x + \frac{1}{10}\sin x + i\left(-\frac{1}{10}\cos x - \frac{3}{10}\sin x\right). \end{aligned}$$

Note that we only require the imaginary part of z_p: thus, a particular solution to the original differential equation is given by

$$y_p = \mathfrak{Im}\left(z_p\right) = -\frac{1}{10}\cos x - \frac{3}{10}\sin x.$$

Therefore, our general solution is

$$y = c_1 e^x + c_2 e^{-2x} - \frac{1}{10}\cos x - \frac{3}{10}\sin x.$$

Example 26. Find the general solution to

$$y'' - 3y' - 4y = 5e^{4x}.$$

Solution. The complementary equation has general solution

$$y_h = c_1 e^{4x} + c_2 e^{-x}.$$

We now find a particular solution: since $f(x) = 5e^{4x}$, we guess $y_p = Ae^{4x}$. Note, however, that such a guess will not work: if we were to construct the general solution $y = y_h + y_p$, this term 'disappears' into y_h, leaving us with no particular solution. We can indeed test this out: plugging $y_p = Ae^{4x}$, $y_p' = 4Ae^{4x}$, and $y_p'' = 16Ae^{4x}$ into the differential equation yields

$$16Ae^{4x} - 3\left(4Ae^{4x}\right) - 4Ae^{4x} = 5e^{4x} \quad \Longrightarrow \quad 0 = 5,$$

which is impossible! We thus *modify* our guess: we try $y_p = Axe^{4x}$:

$$\begin{aligned} y_p' &= Ae^{4x} + 4Axe^{4x} = Ae^{4x}(1 + 4x), \\ y_p'' &= 4Ae^{4x}(1 + 4x) + 4Ae^{4x} = 4Ae^{4x}(2 + 4x). \end{aligned}$$

Substituting these into the original differential equation yields

$$\begin{aligned} 4Ae^{4x}(2 + 4x) - 3\left(Ae^{4x}(1 + 4x)\right) - 4\left(Axe^{4x}\right) &= 5e^{4x} \\ 8A + 16Ax - 3A - 12Ax - 4Ax &= 5. \end{aligned}$$

We find that $5A = 5$, and so our 'new' guess works with $A = 1$. We thus have $y_p = xe^{4x}$, and we have the general solution

$$y = c_1 e^{4x} + c_2 e^{-x} + xe^{4x}.$$

Example 27. Find a general solution for

$$y'' + 4y' + 4y = 6e^{-2x}.$$

Solution. The complementary equation has general solution

$$y_h = c_1 e^{-2x} + c_2 xe^{-2x}.$$

We now find a particular solution for $f(x) = 6e^{-2x}$. The initial guess $y_p = Ae^{-2x}$ does not work, since this term appears in y_h; the modification $y_p = Axe^{-2x}$ also appears in y_h and will not work either. We modify once again: try $y_p = Ax^2e^{-2x}$,

$$\begin{aligned} y_p' &= A\left(2xe^{-2x} - 2x^2e^{-2x}\right) = 2Ae^{-2x}\left(x - x^2\right), \\ y_p'' &= 2A\left[-2e^{-2x}\left(x - x^2\right) + e^{-2x}(1 - 2x)\right] = 2Ae^{-2x}\left(1 - 4x + 2x^2\right). \end{aligned}$$

Substituting these into the original differential equation yields

$$\begin{aligned} 2Ae^{-2x}\left(1 - 4x + 2x^2\right) + 4\left(2Ae^{-2x}\left(x - x^2\right)\right) + 4\left(Ax^2e^{-2x}\right) &= 6e^{-2x} \\ 2A - 8Ax + 4Ax^2 + 8Ax - 8Ax^2 + 4Ax^2 &= 6. \end{aligned}$$

We thus find that $2A = 6$, and our trial solution works with $A = 3$. We thus have $y_p = 3x^2e^{-2x}$, and we obtain the general solution

$$y = c_1 e^{-2x} + c_2 xe^{-2x} + 3x^2e^{-2x}.$$

Example 28. Find a particular solution to

$$y'' + 2y' + 3y = 34e^x \cos 2x.$$

Solution. Observe that

$$f(x) = 34e^x \cos 2x = \mathfrak{Re}\left(34e^{(1+2i)x}\right).$$

Thus, we can instead solve for the real part of the solution to

$$z'' + 2z' + 3z = 34e^{(1+2i)x}.$$

We try $z_p = Ae^{(1+2i)x}$, with

$$z_p' = A(1+2i)e^{(1+2i)x}, \quad z_p'' = A(1+2i)^2 e^{(1+2i)x} = A(-3+4i)e^{(1+2i)x}.$$

Substituting this into the equation, we have

$$A(-3+4i)e^{(1+2i)x} + 2A(1+2i)e^{(1+2i)x} + 3Ae^{(1+2i)x} = 34e^{(1+2i)x},$$

and $A(2+8i) = 34$, or $A = 1-4i$. Thus, a particular solution is given by

$$y_p = \mathfrak{Re}\left((1-4i)e^x(\cos 2x + i\sin 2x)\right) = e^x(\cos 2x + 4\sin 2x).$$

Example 29. Find the general solution to

$$y'' + y = \tan x.$$

Solution. The characteristic equation of the complementary equation is $\lambda^2 + 1 = 0$, with complex roots $\lambda = \pm i$. Thus, we have

$$y_h = c_1 \cos x + c_2 \sin x.$$

Note that the derivatives of the non-homogeneous term $f(x) = \tan x$ do not admit straightforward 'guesses'. We thus produce a particular solution by instead varying the parameters: letting $y_1(x) = \cos x$ and $y_2(x) = \sin x$, a particular solution is given by

$$y_p = u(x)\cos x + v(x)\sin x.$$

We have $y_1' = -\sin x$ and $y_2' = \cos x$, and so

$$\begin{aligned} u(x) &= -\int \frac{y_2 f}{y_1 y_2' - y_1' y_2}\,dx = -\int \frac{\sin x \tan x}{\cos^2 x - (-\sin^2 x)}\,dx = -\int \frac{\sin^2 x}{\cos x}\,dx \\ &= -\int \frac{1-\cos^2 x}{\cos x}\,dx = \sin x - \ln|\sec x + \tan x| + c, \\ v(x) &= \int \frac{y_1 f}{y_1 y_2' - y_1' y_2}\,dx = \int \cos x \tan x\,dx = \int \sin x\,dx = -\cos x + d. \end{aligned}$$

Hence, a particular solution is given by

$$\begin{aligned} y_p &= [\sin x - \ln|\sec x + \tan x|]\cos x - \cos x \sin x \\ &= -\ln|\sec x + \tan x|\cos x. \end{aligned}$$

We thus have the general solution

$$y = c_1 \cos x + c_2 \sin x - \ln|\sec x + \tan x|\cos x.$$

Example 30. Consider the differential equation

$$y'' - \frac{2}{x}y' + \frac{2}{x^2}y = x \ln x.$$

1. Show that two solutions to the complementary homogeneous equation are given by
$$y_1(x) = x, \quad y_2(x) = x^2.$$
This tells us that a general solution to the complementary equation is given by
$$y_h(x) = c_1 x + c_2 x^2.$$

2. Perform a variation of parameters to find a particular solution to the non-homogeneous differential equation.

Solution. We verify that the given functions are solutions to the homogeneous equation

$$y'' - \frac{2}{x}y' + \frac{2}{x^2}y = 0.$$

- For $y_1 = x$, we have the derivatives $y_1' = 1$, $y_1'' = 0$: plugging this into the differential equation yields
$$(0) - \frac{2}{x}(1) + \frac{2}{x^2}(x) = -\frac{2}{x} + \frac{2}{x} = 0.$$

- For $y_2 = x^2$, we have the derivatives $y_2' = 2x$, $y_2'' = 2$: plugging this into the differential equation yields
$$(2) - \frac{2}{x}(2x) + \frac{2}{x^2}\left(x^2\right) = 2 - 4 + 2 = 0.$$

We thus have the general solution:

$$y_h(x) = c_1 x + c_2 x^2.$$

We construct a particular solution by varying the parameters of y_h. We have $f(x) = x \ln x$, $y_1'(x) = 1$ and $y_2'(x) = 2x$, so

$$u(x) = -\int \frac{x^2 \cdot x \ln x}{x \cdot 2x - 1 \cdot x^2}\,dx = -\int x \ln x\,dx = -\frac{x^2}{2}\ln x + \frac{x^2}{4} + c,$$
$$v(x) = \int \frac{x \cdot x \ln x}{x \cdot 2x - 1 \cdot x^2}\,dx = \int \ln x\,dx = x \ln x - x + d.$$

Thus, a particular solution is given by

$$y_p(x) = \left[-\frac{x^2}{2}\ln x + \frac{x^2}{4}\right]x + [x \ln x - x]\,x^2 = \frac{1}{2}x^3 \ln x - \frac{3}{4}x^3.$$

3.2 Simple Harmonic Motion

One of the most important applications of second-order linear differential equations is in modeling the *harmonic oscillator*. Various systems behave in a similar way about an equilibrium: a mass on a spring, a pendulum swinging through a small angle, the charge in a simple circuit, and so on—when pushed slightly, a restoring force pulls them back in proportion to how far they have been displaced.

A particle is said to exhibit **simple harmonic motion** if its acceleration is directly proportional to its displacement:

$$\ddot{x} = -\omega^2 x \quad \Longleftrightarrow \quad \ddot{x} + \omega^2 x = 0.$$

In some cases, a **damping force** may be present: suppose that a resisting force proportional to velocity acts against an oscillating object. The equation of motion then becomes

$$\ddot{x} = -\omega^2 x - 2\gamma\dot{x} \quad \Longleftrightarrow \quad \ddot{x} + 2\gamma\dot{x} + \omega^2 x = 0.$$

We may also be interested in the case where an **external driving force** $f(t)$ independent of the oscillator's motion acts upon the system:

$$\ddot{x} = -\omega^2 x + f(t) \quad \Longleftrightarrow \quad \ddot{x} + \omega^2 x = f(t).$$

In all of these cases, the physics is reduced to a matter of solving a second-order linear differential equation.

Example 31. Find the general solution to the equation of motion of a simple harmonic oscillator:

$$\ddot{x} = -\omega^2 x.$$

Solution. We have a second-order homogeneous linear differential equation, whose characteristic equation is given by

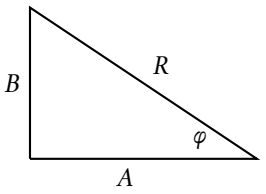
$$\lambda^2 + \omega^2 = 0 \quad \Longrightarrow \quad \lambda = \pm i\omega.$$

A general solution is thus given by

$$x(t) = A\cos\omega t + B\sin\omega t, \quad A, B \in \mathbb{R}.$$

Equivalently, we may write

$$x(t) = R\cos(\omega t - \varphi), \text{ where } R = \sqrt{A^2 + B^2}.$$



To see this, observe that A, B, and $R = \sqrt{A^2 + B^2}$ form the sides of a right triangle, as illustrated on the right. Let φ be the angle between the side of length A and the hypotenuse. Then, $\cos\varphi = A/R$ and $\sin\varphi = B/R$, and we may rewrite our solution as:

$$\begin{aligned} x(t) &= A\cos\omega t + B\sin\omega t = R\left(\frac{A}{R}\cos\omega t + \frac{B}{R}\sin\omega t\right) \\ &= R(\cos\varphi\cos\omega t + \sin\varphi\sin\omega t) = R\cos(\omega t - \varphi). \end{aligned}$$

From this expression of the solution, our physical constants gain meaning: it is now clear why ω is the **natural frequency** of the oscillation. From this, we can compute the **period** of the oscillation to be $T = 2\pi/\omega$. Moreover, we have the **amplitude** of the oscillation, given by R, and the **phase** of the oscillation, given by φ.

Example 32. A particle exhibiting simple harmonic motion starts 5 meters away from equilibrium, with an initial velocity of 0 meters per second. At $t = 0.25$ s, it makes its first return to the initial position. Determine the displacement of the particle as a function of time.

Solution. The position of a harmonic oscillator at time t is given by

$$x(t) = R\cos(\omega t - \varphi).$$

At $t = 0$, we have $x(0) = 5$ and $\dot{x}(0) = 0$: thus,

$$x(0) = R\cos(-\varphi) = R\cos\varphi = 5, \quad \dot{x}(0) = -R\omega\sin(-\varphi) = R\omega\sin\varphi = 0.$$

From these two equations, we have $R = 5$ and $\varphi = 0$, and the equation of motion becomes

$$x(t) = 5\cos(\omega t).$$

The period of one oscillation is 0.25 seconds—that is, at $t = 0.25$, we have

$$x(0.25) = 5\cos(\omega \cdot 0.25) = 5 \implies \omega \cdot 0.25 = 2\pi.$$

Hence, $\omega = 2\pi/0.25 = 8\pi$, and the displacement function of the particle is

$$x(t) = 5\cos(8\pi t).$$

Example 33. The motion of a damped oscillator is described by the differential equation

$$\ddot{x} + 2\gamma\dot{x} + \omega^2 x = 0.$$

Describe the nature of the oscillator's motion for various values of ω and γ.

Solution. The characteristic equation is given by $\lambda^2 + 2\gamma\lambda + \omega^2 = 0$, with solutions

$$\lambda = \frac{-2\gamma \pm \sqrt{4\gamma^2 - 4\omega^2}}{2} = -\gamma \pm \sqrt{\gamma^2 - \omega^2}.$$

The nature of the object's motion thus depends on the value of $\gamma^2 - \omega^2$:

- if $\omega^2 < \gamma^2$, we have real roots $\lambda_1, \lambda_2 \in \mathbb{R}$ and

  $$x(t) = c_1 e^{\lambda_1 t} + c_2 e^{\lambda_2 t},$$

  which describes the motion of an overdamped oscillator,

- if $\omega^2 = \gamma^2$, we have a repeated real root $\lambda \in \mathbb{R}$:

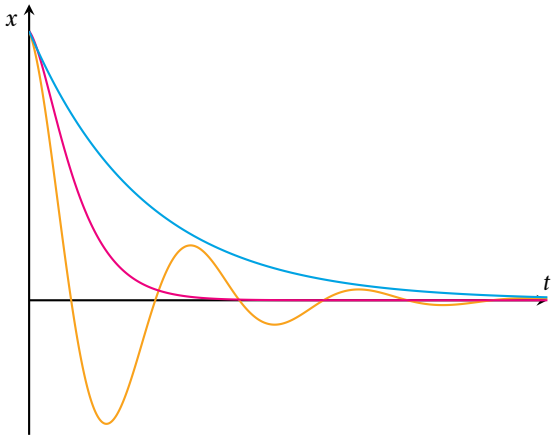
  $$x(t) = c_1 e^{\lambda t} + c_2 t e^{\lambda t},$$

  and we have a critically damped oscillator,

- if $\omega^2 > \gamma^2$, we have complex conjugate roots $\lambda = \alpha + i\beta \in \mathbb{C}$, and

  $$x(t) = c_1 e^{\alpha t}\cos\beta t + c_2 e^{\alpha t}\sin\beta t,$$

  describing the motion of an underdamped oscillator.



Example 34. Consider a horizontal mass-spring system, where an object of mass 0.5 kilograms is attached to a spring with spring constant 5 Newtons per meter. The system is immersed in a medium that provides a resisting force numerically equal to the instantaneous velocity.

1. Write down a differential equation that describes the object's motion. Is the system overdamped, critically damped, or underdamped?
2. Suppose that the mass is initially released from rest, with $x\,(0) = -2$ meters. Determine the function $x\,(t)$ that describes the object's displacement from equilibrium at any time t.

Solution. By Hooke's Law, the object experiences a restoring force $F_{\text{res}} = -kx = -5x$ due to the spring. In addition, the medium imparts a drag force $F_{\text{drag}} = -\dot{x}$. We can then use Newton's Second Law to obtain the equation of motion: the net force on a body is equal to the product of a body's mass with its acceleration, and so

$$0.5\ddot{x} = -5x - \dot{x} \quad \Longrightarrow \quad \ddot{x} + 2\dot{x} + 10x = 0.$$

We obtain the characteristic equation and its roots:

$$\lambda^2 + 2\lambda + 10 = 0 \quad \Longrightarrow \quad \lambda = \frac{-2 \pm \sqrt{2^2 - 4 \cdot 1 \cdot 10}}{2} = -1 \pm 3i.$$

We thus have an underdamped system, whose solution has the form

$$x\,(t) = c_1 e^{-t} \cos 3t + c_2 e^{-t} \sin 3t.$$

From the initial position $x\,(0) = -2$, we find that $x\,(0) = c_1 = -2$. The velocity of the mass is given by

$$\dot{x}\,(t) = -2\left(-e^{-t}\cos 3t - 3e^{-t}\sin 3t\right) + c_2\left(-e^{-t}\sin 3t + 3e^{-t}\cos 3t\right).$$

Since the object starts at rest, we have $\dot{x}\,(0) = 0$, and so

$$\dot{x}\,(0) = 2 + 3c_2 = 0 \quad \Longrightarrow \quad c_2 = -\frac{2}{3}.$$

Hence, the displacement of the mass is given by

$$x\,(t) = -2e^{-t}\cos 3t - \frac{2}{3}e^{-t}\sin 3t.$$

Example 35. Consider an undamped harmonic oscillator equipped with an external sinusoidal driving force, such as one supplied by a motor attached to the oscillator. The equation of motion is given by

$$\ddot{x} = -\omega^2 x + F\cos\psi t.$$

The phenomenon of **resonance** occurs when the angular frequency of the driving force is equal to that of the oscillator: that is, $\psi = \omega$. Determine the function $x(t)$ that describes the oscillator's motion under resonance.

Solution. We wish to solve the non-homogeneous differential equation

$$\ddot{x} + \omega^2 x = F\cos\omega t.$$

We can do this using the method of undetermined coefficients. To begin, we already have the solution to the complementary equation

$$x_h(t) = A\cos\omega t + B\sin\omega t = R\cos(\omega t - \varphi).$$

To find x_p, we consider the non-homogeneous part $F\cos\omega t = \mathfrak{Re}\left\{Fe^{i\omega t}\right\}$. We can thus consider the complex differential equation

$$\ddot{z} + \omega^2 z = Fe^{i\omega t},$$

taking only the real part of its solution. We try $z_p = Ce^{i\omega t}$: note, however, that $\mathfrak{Re}\left\{Ce^{i\omega t}\right\}$ appears in x_h; we thus need to modify our guess. Let $z_p = Cte^{i\omega t}$: then,

$$\begin{aligned}\dot{z}_p &= Ce^{i\omega t} + Ci\omega te^{i\omega t} = Ce^{i\omega t}(1 + i\omega t),\\ \ddot{z}_p &= Ci\omega e^{i\omega t}(1 + i\omega t) + Ci\omega e^{i\omega t} = Ci\omega e^{i\omega t}(2 + i\omega t).\end{aligned}$$

Plugging these into the differential equation yields:

$$Ci\omega e^{i\omega t}(2 + i\omega t) + \omega^2\left(Cte^{i\omega t}\right) = Fe^{i\omega t} \quad\Longrightarrow\quad Ci\omega(2 + i\omega t) + Ct\omega^2 = F.$$

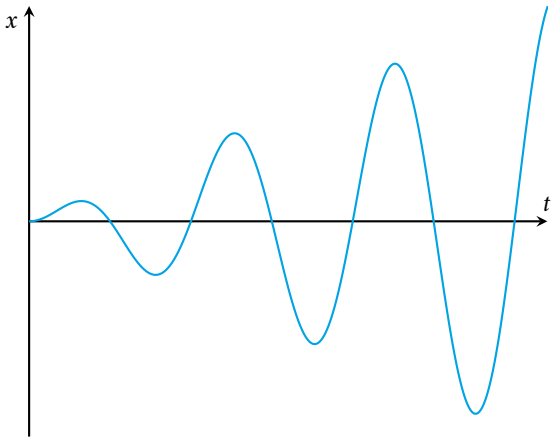
We thus find that $2Ci\omega = F$, or $C = -iF/2\omega$. Our particular solution is thus

$$z_p = \left(-i\frac{F}{2\omega}\right)te^{i\omega t} = -i\frac{F}{2\omega}t(\cos\omega t + i\sin\omega t).$$

We only require the real part of this solution, and thus, the motion of an oscillator under resonance is given by

$$x(t) = R\cos(\omega t - \varphi) + \frac{F}{2\omega}t\sin\omega t.$$

Observe that as t goes on, $x(t)$ oscillates with an amplitude that grows without bound.



4
The Laplace Transform

4.1 Basic Properties

Our methods for solving linear differential equations are no longer sufficient when they involve more complicated non-homogeneous terms (piecewise functions, for instance). Consider a function $f(t)$, viewing the variable $t \geq 0$ as time. The **Laplace transform of $f(t)$** is defined by

$$\mathcal{L}[f(t)] = F(s) = \lim_{h\to\infty} \int_0^h f(t) \cdot e^{-st}\, dt,$$

and $f(t)$ is the **inverse** Laplace transform of $F(s)$:

$$f(t) = \mathcal{L}^{-1}[F(s)].$$

Note that the Laplace transform transforms the function $f(t)$ into a function of a new variable s. We will soon discover that the Laplace transform converts a *differential* equation in t into an *algebraic* equation in s, which allows us to determine solutions by using ordinary algebra.

Example 36. Find the Laplace transform of the function $f(t) = 1$, for $t \geq 0$.

Solution. We directly apply the definition of the Laplace transform:

$$\begin{aligned}\mathcal{L}[1] &= \int_0^\infty e^{-st}\, dt = \lim_{h\to\infty} \left(-\frac{1}{s}e^{-st}\right)\Big|_0^h \\ &= \lim_{h\to\infty} \left(\frac{-e^{-sh}}{s} + \frac{1}{s}\right) = \frac{1}{s}, \quad s > 0.\end{aligned}$$

Example 37. Find the Laplace transform of the function $f(t) = e^{at}$, for $t \geq 0$, where $a \in \mathbb{R}$.

Solution. Applying the definition of the Laplace transform,

$$\begin{aligned}\mathcal{L}\left[e^{at}\right] &= \int_0^\infty e^{at}e^{-st}\, dt = \lim_{h\to\infty} \frac{1}{a-s}e^{(a-s)t}\Big|_0^h \\ &= \lim_{h\to\infty} \left(\frac{1}{a-s}\left(e^{(a-s)h} - 1\right)\right) = \frac{1}{s-a}, \quad s > a.\end{aligned}$$

It is rather cumbersome to compute the Laplace transform of an arbitrary function from scratch. Fortunately, from its definition as an improper integral, the Laplace transform is governed by certain rules* that allow us to more easily find the transforms of products and derivatives of functions.

* See "Properties of the Laplace Transform" in the Appendix for proofs of these results.

Theorem (Linearity). *Given functions $f(t)$ and $g(t)$,*

$$\mathcal{L}[af(t) + bg(t)] = a\mathcal{L}[f(t)] + b\mathcal{L}[g(t)], \text{ for all } a, b \in \mathbb{R}.$$

Theorem (Differentiation Property). *Let $F(s) = \mathcal{L}[f(t)]$. Then,*

$$\mathcal{L}[t^n f(t)] = (-1)^n F^{(n)}(s).$$

Theorem (The First Shifting Theorem). *Let $F(s) = \mathcal{L}[f(t)]$. Then,*

$$\mathcal{L}\left[e^{at} f(t)\right] = F(s-a).$$

Theorem (Transform of Derivatives). *Given a function $f(t)$,*

$$\mathcal{L}\left[f^{(n)}(t)\right] = s^n \mathcal{L}[f(t)] - s^{n-1} f(0) - \cdots - sf^{(n-2)}(0) - f^{(n-1)}(0).$$

Example 38. Find the inverse Laplace transform of $F(s) = \dfrac{1}{s^2 + 2s - 3}$.

Solution. We perform a partial fraction decomposition of $F(s)$:

$$F(s) = \frac{1}{s^2 + 2s - 3} = -\frac{1}{4} \cdot \frac{1}{s+3} + \frac{1}{4} \cdot \frac{1}{s-1}.$$

We take the inverse transform of $F(s)$ term by term, with reference to the table of Laplace transforms:

$$\mathcal{L}^{-1}[F(s)] = -\frac{1}{4} \cdot \mathcal{L}^{-1}\left[\frac{1}{s+3}\right] + \frac{1}{4} \cdot \mathcal{L}^{-1}\left[\frac{1}{s-1}\right] = -\frac{1}{4}e^{-3t} + \frac{1}{4}e^{t}.$$

Example 39. Compute $\mathcal{L}[f(t)]$ for the function $f(t) = t$, $t \geq 0$.

Solution. Applying the definition of the Laplace transform,

$$\begin{aligned}\mathcal{L}[t] &= \lim_{h\to\infty} \int_0^h t \cdot e^{-st}\, dt = \lim_{h\to\infty} \left[e^{-st}\left(-\frac{t}{s} - \frac{1}{s^2}\right)\right]\Bigg|_{t=0}^{t=h} \\ &= \lim_{h\to\infty} \left(-\frac{he^{-sh}}{s} - \frac{e^{-sh}}{s^2} + \frac{1}{s^2}\right) = \frac{1}{s^2}, \quad s > 0.\end{aligned}$$

Alternatively, we could have more easily computed this using the Differentiation Property: start with

$$f(t) = 1 \quad \Longrightarrow \quad \mathcal{L}[1] = \frac{1}{s}.$$

Then,

$$\mathcal{L}[t] = \mathcal{L}[t \cdot 1] = -\frac{d}{ds}\left(\frac{1}{s}\right) = -\left(-\frac{1}{s^2}\right) = \frac{1}{s^2}.$$

Example 40. Evaluate $\mathcal{L}\left[t^2\right]$ and $\mathcal{L}\left[t^3\right]$.

Solution. Evaluating the Laplace transform of $f(t) = t^2$ by directly applying the definition will be rather tedious. We instead apply the Differentiation Property: since $f(t) = t \cdot t$, we have

$$\mathcal{L}\left[t^2\right] = \mathcal{L}[t \cdot t] = -\frac{d}{ds}\mathcal{L}[t] = -\frac{d}{ds}\left(\frac{1}{s^2}\right) = \frac{2}{s^3}.$$

We can repeat this process to evaluate the Laplace transform of $f(t) = t^3$:

$$\mathcal{L}\left[t^3\right] = \mathcal{L}\left[t \cdot t^2\right] = -\frac{d}{ds}\mathcal{L}\left[t^2\right] = -\frac{d}{ds}\left(\frac{2}{s^3}\right) = \frac{3 \cdot 2}{s^4}.$$

Example 41. Evaluate $\mathcal{L}\left[e^{at}\cos\omega t\right]$.

Solution. Let $F(s)$ be the Laplace transform of $\cos\omega t$:

$$F(s) = \mathcal{L}\left[\cos\omega t\right] = \frac{s}{s^2+\omega^2}.$$

By the First Shifting Theorem, we simply replace every instance of s in $F(s)$ with $s-a$:

$$\mathcal{L}\left[e^{at}\cos\omega t\right] = F(s-a) = \frac{s-a}{(s-a)^2+\omega^2}.$$

Example 42. Find the inverse Laplace transform of

$$F(s) = \frac{1-3s}{s^2+8s+32}.$$

Solution. Observe that the denominator cannot be factored; thus, we try a different way of manipulating this expression. Completing the square,

$$s^2+8s+32 = \left(s^2+8s+4^2\right)+32-4^2 = (s+4)^2+16.$$

We may thus apply the First Shifting Theorem, but in order to do so, we need to rewrite the numerator.

$$\begin{aligned} F(s) &= \frac{1-3(s+4-4)}{(s+4)^2+16} = \frac{-3(s+4)+13}{(s+4)^2+16} \\ &= -\frac{3(s+4)}{(s+4)^2+4^2} + \frac{13}{(s+4)^2+4^2}. \end{aligned}$$

We now take the inverse Laplace transform of $F(s)$, observing that these functions look precisely like the Laplace transforms of $\cos 4t$ and $\sin 4t$, with s replaced by $s+4$:

$$\begin{aligned} \mathcal{L}^{-1}\left[F(s)\right] &= -3\mathcal{L}^{-1}\left[\frac{s+4}{(s+4)^2+4^2}\right] + \frac{13}{4}\mathcal{L}^{-1}\left[\frac{4}{(s+4)^2+4^2}\right] \\ &= -3e^{-4t}\cos 4t + \frac{13}{4}e^{-4t}\sin 4t. \end{aligned}$$

Example 43. Solve the initial value problem

$$y' + 2y = 0, \quad y(0) = 2.$$

Solution. Taking the Laplace transform of both sides of the differential equation, we have

$$\mathcal{L}\left[y'+2y\right] = \mathcal{L}\left[0\right].$$

Evaluating the left-hand side, observe that

$$\begin{aligned} \mathcal{L}\left[y'+2y\right] &= \left(s\mathcal{L}\left[y\right] - y(0)\right) + 2\mathcal{L}\left[y\right] \\ &= \left(s\mathcal{L}\left[y\right] - 2\right) + 2\mathcal{L}\left[y\right] \\ &= (s+2)\,\mathcal{L}\left[y\right] - 2. \end{aligned}$$

The right-hand side is simply $\mathcal{L}\left[0\right] = 0$. Thus, we have

$$(s+2)\,\mathcal{L}\left[y\right] - 2 = 0 \quad \Longrightarrow \quad \mathcal{L}\left[y\right] = \frac{2}{s+2}.$$

Therefore, taking the inverse Laplace transform of both sides, we arrive at the particular solution to the differential equation

$$y = 2\cdot\mathcal{L}^{-1}\left[\frac{1}{s+2}\right] = 2e^{-2t}.$$

Notice that the differential equation is separable, and we would have arrived at the same answer by performing a separation of variables!

Example 44. Use the Laplace transform to solve the initial value problem

$$y'' - y' - 2y = 0, \quad y(0) = 1, \quad y'(0) = 0.$$

Solution. We take the Laplace transform of both sides of the equation:

$$\mathcal{L}[y'' - y' - 2y] = \mathcal{L}[0].$$

We expand the left-hand side of the equation:

$$\begin{aligned}\mathcal{L}[y'' - y' - 2y] &= \mathcal{L}[y''] - \mathcal{L}[y'] - 2\mathcal{L}[y] \\ &= \left(s^2\mathcal{L}[y] - sy(0) - y'(0)\right) - (s\mathcal{L}[y] - y(0)) - 2\mathcal{L}[y] \\ &= \left(s^2\mathcal{L}[y] - s\right) - (s\mathcal{L}[y] - 1) - 2\mathcal{L}[y] \\ &= \left(s^2 - s - 2\right)\mathcal{L}[y] - s + 1.\end{aligned}$$

Since $\mathcal{L}[0] = 0$, we have

$$\left(s^2 - s - 2\right)\mathcal{L}[y] - s + 1 = 0 \quad \Longrightarrow \quad \mathcal{L}[y] = \frac{s-1}{s^2 - s - 2} = \frac{s-1}{(s-2)(s+1)}.$$

To compute the inverse Laplace transform of $\mathcal{L}[y]$, we take its partial fraction decomposition:

$$\frac{s-1}{(s-2)(s+1)} = \frac{A}{s-2} + \frac{B}{s+1} = \frac{(A+B)s + A - 2B}{s^2 - s - 2}.$$

Thus, $A + B = 1$ and $A - 2B = -1$. Hence, $B = 2/3$ and $A = 1/3$, and

$$\mathcal{L}[y] = \frac{s-1}{(s-2)(s+1)} = \frac{1}{3}\cdot\frac{1}{s-2} + \frac{2}{3}\cdot\frac{1}{s+1}.$$

Taking the inverse Laplace transform of both sides, we find that

$$y = \mathcal{L}^{-1}\left[\frac{1}{3}\cdot\frac{1}{s-2} + \frac{2}{3}\cdot\frac{1}{s+1}\right] = \frac{1}{3}e^{2t} + \frac{2}{3}e^{-t}.$$

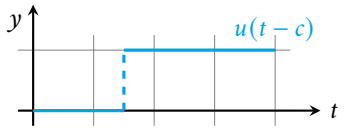
Verify that you obtain the same result by solving the equation as a homogeneous linear equation!

4.2 Step Functions and the Unit Impulse

In practice, many engineering systems will involve functions that are not particularly simple: a motor might only run for a finite amount of time before it shuts off, or a hammer might strike an object at a single instant. The Laplace transform now allows us to solve differential equations that might involve these less 'tidy' functions.

The simplest example of a *piecewise function* is the **unit step function**, denoted by $u(t-c)$, and defined by

$$u(t-c) = \begin{cases} 0, & t < c, \\ 1, & t > c. \end{cases}$$



Observe that multiplying a function $f(t)$ by a unit step allows us to switch a function 'on' and 'off': the following theorem thus allows us to easily compute the Laplace transform of a piecewise function.

Theorem (The Second Shifting Theorem)**.** *Let* $F(s) = \mathcal{L}[f(t)]$. *Then,*

$$\mathcal{L}[f(t-c)\,u(t-c)] = e^{-sc}F(s).$$

Equivalently, we may write

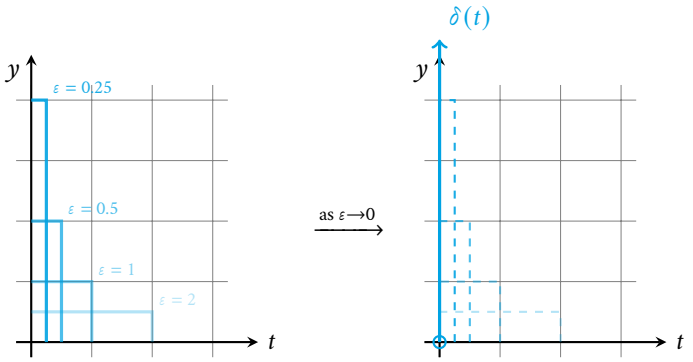
$$\mathcal{L}[f(t)\,u(t-c)] = e^{-sc}\mathcal{L}[f(t+c)].$$

We now characterize *instantaneous phenomena*: consider the function

$$f(t) = \begin{cases} 1/\varepsilon, & 0 < t < \varepsilon, \\ 0, & t > \varepsilon. \end{cases}$$

The **Dirac delta function** (or the **unit impulse function**) is defined by taking the limit of $f(t)$ as $\varepsilon \to 0$:

$$\delta(t) = \lim_{\varepsilon \to 0} f(t).$$



Its Laplace transform is given by

$$\mathcal{L}[\delta(t-c)] = e^{-cs}.$$

Example 45. Use the definition of the Laplace transform to compute $\mathcal{L}[u(t-c)]$ for some fixed $c \geq 0$.

Solution. We apply the definition of the Laplace transform:

$$\mathcal{L}[u(t-c)] = \lim_{h\to\infty} \int_0^h e^{-st} u(t-c)\, dt.$$

Since $u(t-c)$ is a piecewise function, we may split the integral over the domains $t \in (0,c)$, where $u(t-c) = 0$, and $t \in (c,h)$, where $u(t-c) = 1$:

$$\begin{aligned}
\mathcal{L}[u(t-c)] &= \int_0^c e^{-st} u(t-c)\, dt + \lim_{h\to\infty} \int_c^h e^{-st} u(t-c)\, dt \\
&= \int_0^c e^{-st} \cdot 0\, dt + \lim_{h\to\infty} \int_c^h e^{-st} \cdot 1\, dt \\
&= \lim_{h\to\infty} \left[-\frac{e^{-st}}{s} \right]\Bigg|_{t=c}^{t=h} = \lim_{h\to\infty} \left[\frac{e^{-sc}}{s} - \frac{e^{-sh}}{s} \right] = \frac{e^{-cs}}{s}.
\end{aligned}$$

Example 46. Evaluate $\mathcal{L}\left[e^{2t} \cdot u(t-4)\right]$.

Solution. We apply the Second Shifting Theorem:

$$\mathcal{L}\left[e^{2t} \cdot u(t-4)\right] = e^{-4s}\mathcal{L}\left[e^{2(t+4)}\right] = e^{-4s}\mathcal{L}\left[e^{2t}e^{8}\right] = \frac{e^{-4s+8}}{s-2}.$$

Example 47. Evaluate $\mathcal{L}^{-1}\left[\left(1-e^{-2s}\right)s^{-2}\right]$.

Solution. The exponential term suggests that $f(t) = \mathcal{L}^{-1}[F(s)]$ involves the unit step. We apply the Second Shifting Theorem:

$$\begin{aligned}
\mathcal{L}^{-1}\left[\frac{1-e^{-2s}}{s^2}\right] &= \mathcal{L}^{-1}\left[\frac{1}{s^2}\right] - \mathcal{L}^{-1}\left[\frac{e^{-2s}}{s^2}\right] \\
&= t - (t-2)\cdot u(t-2).
\end{aligned}$$

We may express this as a piecewise function:

$$f(t) = \begin{cases} t, & 0 \leq t < 2, \\ 2, & t > 2. \end{cases}$$

Example 48. Find the inverse Laplace transform of

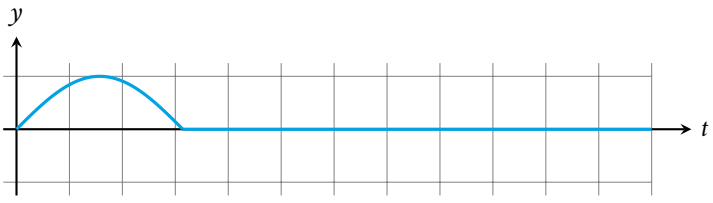
$$F(s) = \frac{1+e^{-\pi s}}{s^2+1}.$$

Solution. Applying the Second Shifting Theorem,

$$\begin{aligned}
\mathcal{L}^{-1}\left[\frac{1+e^{-\pi s}}{s^2+1}\right] &= \mathcal{L}^{-1}\left[\frac{1}{s^2+1}\right] + \mathcal{L}^{-1}\left[\frac{e^{-\pi s}}{s^2+1}\right] \\
&= \sin t + u(t-\pi)\sin(t-\pi).
\end{aligned}$$

Note that since $\sin(t-\pi) = -\sin t$, we have

$$f(t) = \sin t\,(1-u(t-\pi)) = \begin{cases} \sin t, & 0 < t < \pi, \\ 0, & t > \pi. \end{cases}$$



Example 49. Describe the graph of

$$y(t) = u(t-a) - u(t-b), \text{ where } 0 \le a < b.$$

Hence, describe the graph of

$$z(t) = f(t)\left[u(t-a) - u(t-b)\right], \text{ where } 0 \le a < b.$$

Solution. By the definition of the unit step function,

$$y(t) = \begin{cases} 0 - 0 = 0, & t < a, \\ 1 - 0 = 1, & a < t < b, \\ 1 - 1 = 0, & t > b. \end{cases}$$

This is simply a function that has value 1 over the interval (a, b) and is zero elsewhere. Thus, the function $z(t)$ takes on the graph of $f(t)$ over the interval (a, b) and zero elsewhere:

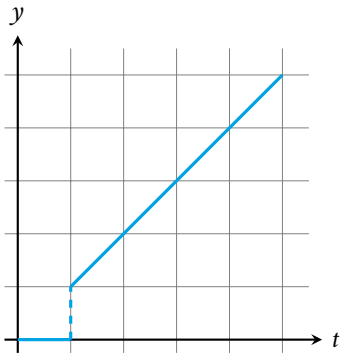
$$z(t) = \begin{cases} 0, & t < a, \\ f(t), & a < t < b, \\ 0, & t > b. \end{cases}$$

Example 50. Let $f(t) = t \cdot u(t-1)$.

1. Sketch the graph of $f(t)$.
2. Evaluate $\mathcal{L}[f(t)]$.

Solution. We may express $f(t)$ as a piecewise function:

$$f(t) = \begin{cases} 0 & \text{when } t < 1, \\ t & \text{when } t > 1. \end{cases}$$



We directly apply the Second Shifting Theorem:

$$\mathcal{L}[t \cdot u(t-1)] = e^{-s}\mathcal{L}[t+1] = e^{-s}\left(\frac{1}{s^2} + \frac{1}{s}\right) = \frac{1+s}{s^2 e^s}.$$

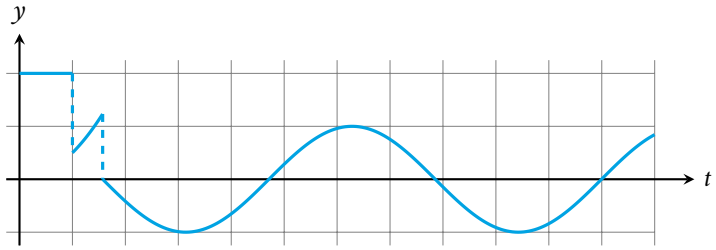
Alternatively, we can apply the Differentiation Property:

$$\mathcal{L}[t \cdot u(t-1)] = -\frac{d}{ds}\mathcal{L}[u(t-1)] = -\frac{d}{ds}\left(\frac{e^{-s}}{s}\right) = \frac{1+s}{s^2 e^s}.$$

Example 51. Consider the piecewise function given by

$$f(t) = \begin{cases} 2, & 0 < t < 1, \\ t^2/2, & 1 < t < \pi/2, \\ \cos t, & t > \pi/2. \end{cases}$$

Write $f(t)$ in terms of the unit step function, and hence find the Laplace transform of $f(t)$.



Solution. We can rewrite $f(t)$ by taking the products of the function's values with the corresponding difference of unit step functions:

$$\begin{aligned} f(t) &= 2(1 - u(t-1)) + \frac{t^2}{2}\left[u(t-1) - u\left(t - \frac{\pi}{2}\right)\right] + \cos t \cdot u\left(t - \frac{\pi}{2}\right) \\ &= 2(1 - u(t-1)) + \frac{t^2}{2} \cdot u(t-1) - \frac{t^2}{2} \cdot u\left(t - \frac{\pi}{2}\right) + \cos t \cdot u\left(t - \frac{\pi}{2}\right). \end{aligned}$$

We now take the Laplace transforms of each term. The first term is fairly straightforward:

$$\mathcal{L}[2(1 - u(t-1))] = 2\left(\frac{1}{s} - \frac{e^{-s}}{s}\right).$$

For the second term, we apply the Second Shifting Theorem,

$$\begin{aligned} \mathcal{L}\left[\frac{t^2}{2} \cdot u(t-1)\right] &= \frac{1}{2}e^{-s}\mathcal{L}\left[(t+1)^2\right] = \frac{e^{-s}}{2}\mathcal{L}\left[t^2 + 2t + 1\right] \\ &= \frac{e^{-s}}{2}\left(\frac{2!}{s^3} + \frac{2}{s^2} + \frac{1}{s}\right). \end{aligned}$$

Similarly for the third term,

$$\begin{aligned} \mathcal{L}\left[\frac{t^2}{2} \cdot u\left(t - \frac{\pi}{2}\right)\right] &= \frac{1}{2}e^{-s\pi/2}\mathcal{L}\left[\left(t + \frac{\pi}{2}\right)^2\right] = \frac{1}{2}e^{-s\pi/2}\mathcal{L}\left[t^2 + \pi t + \frac{\pi^2}{4}\right] \\ &= \frac{1}{2}e^{-s\pi/2}\left(\frac{2!}{s^3} + \frac{\pi}{s^2} + \frac{\pi^2}{4s}\right). \end{aligned}$$

Finally, we have

$$\begin{aligned} \mathcal{L}\left[\cos t \cdot u\left(t - \frac{\pi}{2}\right)\right] &= e^{-s\pi/2}\mathcal{L}\left[\cos\left(t + \frac{\pi}{2}\right)\right] = e^{-s\pi/2}\mathcal{L}[-\sin t] \\ &= -\frac{e^{-s\pi/2}}{s^2 + 1}. \end{aligned}$$

We can now combine all our results to form $F(s) = \mathcal{L}[f(t)]$:

$$F(s) = 2\left(\frac{1 - e^{-s}}{s}\right) + e^{-s}\left(\frac{1}{s^3} + \frac{1}{s^2} + \frac{1}{2s}\right) - e^{-s\pi/2}\left(\frac{1}{s^3} + \frac{\pi}{2s^2} + \frac{\pi^2}{8s} + \frac{1}{s^2 + 1}\right).$$

Example 52. Consider a damped oscillator which is struck by a hammer at time $t = 1$. The equation of motion is given by the differential equation

$$\ddot{y} + 3\dot{y} + 2y = \delta(t-1), \quad y(0) = \dot{y}(0) = 0.$$

Determine the function $y(t)$ that describes the motion of the oscillator.

Solution. Taking the Laplace transform of both sides, we have

$$s^2\mathcal{L}[y] + 3s\mathcal{L}[y] + 2\mathcal{L}[y] = e^{-s} \quad \Longrightarrow \quad \mathcal{L}[y] = \frac{e^{-s}}{s^2+3s+2}.$$

Observe that we may rewrite the above expression—taking a partial fraction decomposition

$$\frac{e^{-s}}{s^2+3s+2} = \frac{e^{-s}}{(s+1)(s+2)} = \left(\frac{1}{s+1} - \frac{1}{s+2}\right)e^{-s}.$$

Taking the inverse of the above function,

$$\mathcal{L}^{-1}\left[\left(\frac{1}{s+1} - \frac{1}{s+2}\right)e^{-s}\right] = u(t-1)\left(e^{-(t-1)} - e^{-2(t-1)}\right).$$

Thus, the solution to the initial value problem is given by

$$y(t) = \begin{cases} 0, & 0 < t < 1, \\ e^{-(t-1)} - e^{-2(t-1)}, & t > 1. \end{cases}$$

5
An Introduction to Partial Differential Equations

5.1 First Principles

Many physical quantities depend on more than a single variable, and we may be interested in how a function changes with respect to each of these independent variables. A **multivariable function** f is a rule that assigns every n-tuple of real numbers $(x_1, \ldots, x_n)$ a unique real number $f(x_1, \ldots, x_n)$.

For simplicity, consider a function of two variables, $f(x, y)$. We can investigate how f changes with respect to x, as well as with respect to y: the **partial derivative of f with respect to x** is the derivative of the function $g(x) = f(x, y)$, where y is considered a constant, and the **partial derivative of $f(x, y)$ with respect to y** is the derivative of the function $h(y) = f(x, y)$, where x is considered a constant. We respectively denote these as any of the following expressions:

$$\frac{\partial f}{\partial x} = f_x, \qquad \frac{\partial f}{\partial y} = f_y.$$

As with ordinary derivatives, we can take **higher-order partial derivatives** by differentiating the partial derivatives:

$$\frac{\partial^2 f}{\partial x^2} = f_{xx}, \quad \frac{\partial^2 f}{\partial y^2} = f_{yy}, \quad \frac{\partial^2 f}{\partial x \partial y} = f_{xy}, \quad \frac{\partial^2 f}{\partial y \partial x} = f_{yx}, \quad \ldots.$$

A **partial differential equation** is an equation involving one or more partial derivatives of a multivariable function. The **order** of a partial differential equation is the order of the highest derivative in the equation, and a **solution** of a partial differential equation is a function that satisfies the equation.

Example 53. Consider the partial differential equation

$$au_x + bu_y = 0, \text{ for some } a, b \in \mathbb{R}.$$

Show that a solution to the above equation is given by $u(x, y) = f(bx - ay)$, where $f(x)$ is any function of a single variable.

Solution. We simply evaluate the partial derivatives of $u(x, y)$ and show that they satisfy the given equation. We employ the Chain Rule:

$$u_x(x, y) = f'(bx - ay) \cdot b, \quad u_y = f'(bx - ay) \cdot (-a).$$

Thus,

$$au_x + bu_y = a(bf') + b(-af') = 0,$$

and indeed $u(x, y)$ is a solution to the partial differential equation.

Example 54. Determine a solution $u(x, y)$ to the second-order partial differential equation

$$u_{xx} = 0.$$

Solution. We can simply integrate both sides of the equation twice with respect to x. Note that our arbitrary constants may be functions of y: integrating once yields

$$\int u_{xx}\, dx = \int 0\, dx \quad \Longrightarrow \quad u_x = f(y).$$

Integrating once again, we have

$$\int u_x\, dx = \int f(y)\, dx \quad \Longrightarrow \quad u(x, y) = f(y)\, x + g(y).$$

Note that there are two arbitrary *functions* in this solution.

If the equation is of the first degree in the (unknown) function and its derivatives, then it is said to be **linear**; in addition, if each of the terms in the linear equation contains either the unknown function or one of its partial derivatives, then the equation is **homogeneous**. In general, there are many solutions to a single partial differential equation, and unlike ordinary differential equations, these solutions may look entirely different from one another.

Theorem (Superposition Principle)**.** *Let $u_1(x, y)$ and $u_2(x, y)$ be solutions of a homogeneous linear partial differential equation. Then, a solution is also given by*

$$u(x, y) = c_1 u_1(x, y) + c_2 u_2(x, y), \text{ for any } c_1, c_2 \in \mathbb{R}.$$

Example 55. The two-dimensional Laplace equation is given by

$$u_{xx} + u_{yy} = 0.$$

Show that the functions $u_1(x, y) = x^2 - y^2$ and $u_2(x, y) = e^x \cos y$ are both solutions to the Laplace equation. Hence, show that a solution to the Laplace equation is also given by

$$u(x, y) = 5x^2 - 5\left(e^x \cos y + y^2\right).$$

Solution. We compute the partial derivatives of $u_1(x, y) = x^2 - y^2$:

$$u_{xx} = (2x)_x = 2, \quad u_{yy} = (-2y)_y = -2.$$

Plugging this into the equation, we find that $u_1(x, y)$ is indeed a solution:

$$u_{xx} + u_{yy} = 2 - 2 = 0.$$

Similarly, we compute the partial derivatives of $u_2(x, y) = e^x \cos y$:

$$u_{xx} = (e^x \cos y)_x = e^x \cos y, \quad u_{yy} = (-e^x \sin y)_y = -e^x \cos y,$$

and we verify that $u_2(x, y)$ satisfies the differential equation:

$$u_{xx} + u_{yy} = e^x \cos y - e^x \cos y = 0.$$

By the Superposition Principle, we find that

$$u(x, y) = 5x^2 - 5\left(e^x \cos y + y^2\right) = 5\left(x^2 - y^2\right) - 5e^x \cos y$$

must also be a solution to the partial differential equation.

Given a homogeneous, linear partial differential equation, we can 'construct' a solution by the **method of separation of variables**:

1. Suppose that a solution exists of the form $u(x, y) = X(x)\,Y(y)$.
2. Replace all expressions of u (e.g., $u, u_x, u_y, \dots$) in the differential equation with their corresponding expression in terms of X and Y (e.g., $u = XY$, $u_x = X'Y, u_y = XY', u_{xy} = X'Y', \dots$).
3. Separate all the terms involving x and the terms involving y, and equate both sides to a *separation constant* k.
4. Solve the ordinary differential equations in x and y to obtain $u(x, y) = X(x)\,Y(y)$.

Example 56. Use the method of separation of variables to find a solution to

$$u_x + xu_y = 0.$$

Solution. Suppose that a solution of the form $u(x, y) = X(x)\,Y(y)$ exists. Then,

$$u_x(x, y) = X'(x)\,Y(y), \quad u_y(x, y) = X(x)\,Y'(y).$$

Plugging this into the differential equation, we find that

$$X'Y + xXY' = 0.$$

We now perform a separation of variables, noting that the functions X and X' only involve the variable x, and Y, Y' only involve the variable y:

$$X'Y = -xXY' \quad \Longrightarrow \quad \frac{1}{xX} \cdot \frac{dX}{dx} = -\frac{1}{Y} \cdot \frac{dY}{dy}.$$

Observe that equality holds for all values of x and y—this is only possible if both expressions are constant. Thus, we may set

$$\frac{1}{xX} \cdot \frac{dX}{dx} = -\frac{1}{Y} \cdot \frac{dY}{dy} = k \text{ for some } k \in \mathbb{R}.$$

This gives rise to two ordinary differential equations, one in x, and another in y:

$$\frac{1}{xX} \cdot \frac{dX}{dx} = k, \quad -\frac{1}{Y} \cdot \frac{dY}{dy} = k.$$

We solve these as separable ordinary differential equations:

$$\int \frac{1}{X}\,dX = \int kx\,dx \quad \Longrightarrow \quad \ln|X| = k\frac{x^2}{2} + c.$$

Exponentiating both sides yields the function

$$X(x) = e^{kx^2/2+c} = Ae^{kx^2/2}.$$

Similarly, we have

$$\int \frac{1}{Y}\,dY = \int -k\,dy \quad \Longrightarrow \quad \ln|Y| = -ky + d.$$

Exponentiating both sides yields

$$Y(y) = e^{-ky+d} = Be^{-ky}.$$

By our initial assumption, the product of the two functions $X(x)$ and $Y(y)$ forms the solution to the original differential equation:

$$u(x, y) = X(x)\,Y(y) = \left(Ae^{kx^2/2}Be^{-ky}\right) = Ce^{k\left(x^2/2-y\right)}.$$

Example 57. Use the method of separation of variables to find a solution to the partial differential equation

$$xu_x = u + yu_y$$

that satisfies the conditions $u(1,1) = 2$ and $u(1,2) = 8$.

Solution. Let $u = X(x)\,Y(y)$. Then, $u_x = X'Y$ and $u_y = XY'$, and the differential equation becomes

$$xX'Y = XY + yXY' \quad \Longrightarrow \quad \frac{x}{X}X' - 1 = \frac{y}{Y}Y'.$$

Setting both sides equal to k, we arrive at two ordinary differential equations which we can solve:

$$\frac{x}{X}X' - 1 = k, \quad \frac{y}{Y}Y' = k.$$

We solve both of these equations in turn by performing a standard separation of variables:

$$\int \frac{1}{X}\,dX = \int (k+1)\,\frac{1}{x}\,dx \implies \ln|X| = (k+1)\ln|x| + c,$$

$$\int \frac{1}{Y}\,dY = \int k\frac{1}{y}\,dy \implies \ln|Y| = k\ln|y| + d.$$

Exponentiating both sides, we thus find that

$$X(x) = Ax^{k+1}, \quad Y(y) = By^k.$$

So, a solution to the partial differential equation is given by

$$u(x,y) = X(x)\,Y(y) = Cx^{k+1}y^k, \text{ where } C \in \mathbb{R}.$$

To find a particular solution, we use the given conditions:

$$\begin{aligned} u(1,1) = C \cdot 1 \cdot 1 = 2 \quad &\Longrightarrow \quad C = 2, \\ u(1,2) = 2 \cdot 1 \cdot 2^k = 8 \quad &\Longrightarrow \quad k = 2. \end{aligned}$$

Thus, the particular solution is given by

$$u(x,y) = 2x^3y^2.$$

5.2 The Heat Equation

Let us now apply the method of separation of variables to investigate one of the most important partial differential equations in engineering. Consider a thin metal rod of length ℓ, insulated along its sides so that heat can only flow along its length. Let $u(x,t)$ be the temperature of the rod at a given position x and time t: as time passes, heat flows from hotter regions to cooler ones. This dispersion of heat is modeled by the **heat equation**

$$u_t = c^2 u_{xx}, \quad 0 < x < \ell, \quad t > 0.$$

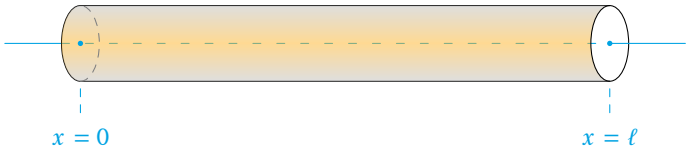
The constant c^2 is the *thermal diffusivity* of the metal, which measures how quickly the material conducts heat.

To determine a particular solution, we require additional information. Suppose that, at its two ends $x = 0$ and $x = \ell$, the rod is held at a constant temperature: setting thermal equilibrium at 0, we impose the **boundary conditions**

$$u(0,t) = 0, \qquad u(\ell,t) = 0.$$

If the initial distribution of heat at time $t = 0$ is given by some function $f(x)$, then we impose the **initial condition**

$$u(x,0) = f(x).$$



Example 58. Use the method of separation of variables to find a solution to the heat equation given the initial and boundary conditions

$$u_t = c^2 u_{xx}, \quad u(0,t) = u(\ell,t) = 0, \quad u(x,0) = f(x).$$

Solution. Let $u(x,t) = X(x)\,T(t)$. Then, $u_t = XT'$, $u_{xx} = X''T$: plugging this into the heat equation,

$$XT' = c^2 X''T \quad \Longrightarrow \quad \frac{1}{c^2}\cdot\frac{1}{T}\frac{dT}{dt} = \frac{1}{X}\frac{d^2X}{dx^2}.$$

Setting both sides equal to a constant k, we obtain two ordinary differential equations. The first-order differential equation in t is easy to solve:

$$\int \frac{1}{T}\,dT = \int c^2 k\,dt \quad \Longrightarrow \quad \ln|T| = c^2 kt + d \quad \Longrightarrow \quad T = Ae^{c^2 kt}.$$

Now, consider the second-order differential equation in x:

$$\frac{1}{X}\frac{d^2X}{dx^2} = k \quad \Longrightarrow \quad X'' - kX = 0.$$

This is a second-order homogeneous linear differential equation: the characteristic equation is given by

$$\lambda^2 - k = 0 \quad \Longrightarrow \quad \lambda = \pm\sqrt{k}.$$

The nature of the solution $X(x)$ now depends on the value of k, but we know that it will involve either exponentials or sines and cosines.

By our boundary conditions, $u(x,t)$ must be 0 at both $x = 0$ and $x = \ell$. This would be impossible if $u(x,t)$ only involved exponential terms; hence, $X(x)$ must be trigonometric. In particular, we require that $k < 0$, which yields the solution

$$X(x) = c_1 \cos\left(\sqrt{-k}x\right) + c_2 \sin\left(\sqrt{-k}x\right).$$

Thus, a solution to the heat equation is given by

$$u(x,t) = X(x)\,T(t) = e^{c^2kt}\left(\alpha\cos\left(\sqrt{-k}x\right) + \beta\sin\left(\sqrt{-k}x\right)\right).$$

We now find the values of the constants α, β, and k. Given the boundary conditions $u(0,t) = 0$ and $u(\ell,t) = 0$,

$$u(0,t) = e^{c^2kt}(\alpha\cos 0 + \beta\sin 0) = \alpha e^{c^2kt} = 0 \quad\Longrightarrow\quad \alpha = 0,$$

$$u(\ell,t) = e^{c^2kt}\left(\beta\sin\left(\ell\sqrt{-k}\right)\right) = 0 \quad\Longrightarrow\quad \beta\sin\left(\ell\sqrt{-k}\right) = 0.$$

Since α and β cannot both be 0, the second equation implies that

$$\sin\left(\ell\sqrt{-k}\right) = 0 \quad\Longrightarrow\quad \sqrt{-k} = \frac{n\pi}{\ell},\ \text{for } n = 1,2,3,\ldots.$$

Thus, $k = -n^2\pi^2/\ell^2$. Thus, our solution is now given by

$$u(x,t) = \beta_n e^{-c^2n^2\pi^2t/\ell^2}\sin\left(\frac{n\pi}{\ell}x\right),\ \text{for } n = 1,2,3,\ldots.$$

By the Superposition Principle, we can construct the series solution

$$u(x,t) = \sum_{n=1}^{\infty}\beta_n e^{-c^2n^2\pi^2t/\ell^2}\sin\left(\frac{n\pi}{\ell}x\right).$$

In order to deduce each β_n, we use the initial condition $u(x,0) = f(x)$:

$$u(x,0) = \sum_{n=1}^{\infty}\beta_n\sin\left(\frac{n\pi}{\ell}x\right) = f(x).$$

Observe that for different functions $f(x)$, the heat equation will have different terms with different coefficients and, as expected, different particular solutions.

Example 59. Solve the partial differential equation

$$u_t = 2u_{xx}, \qquad 0 < x < 3, \qquad t > 0,$$

given the boundary conditions $u(0,t) = u(3,t) = 0$ and the initial condition $u(x,0) = 5\sin(4\pi x)$.

Solution. This is simply the heat equation with $c^2 = 2$, $\ell = 3$, and $f(x) = 5\sin(4\pi x)$. We have already found its solution:

$$u(x,t) = \beta_n e^{-2n^2\pi^2t/9}\sin\left(\frac{n\pi}{3}x\right),$$

and it remains to determine the values of n and β_n. By the initial condition $u(x,0) = 5\sin(4\pi x)$, we have at $t = 0$

$$\beta_n\sin\left(\frac{n\pi}{3}x\right) = 5\sin(4\pi x).$$

Observe that n must hence be 12, and so $\beta_{12} = 5$. Thus, a solution to the heat equation under these conditions is given by

$$u(x,t) = 5e^{-2\cdot 144\cdot\pi^2t/9}\sin(4\pi x) = 5e^{-32\pi^2t}\sin(4\pi x).$$

Appendix

A Note on the Variation of Parameters

Consider a non-homogeneous, second-order linear differential equation

$$y'' + py' + qy = f,$$

where p, q, and f are some known continuous functions of x. Suppose that the complementary homogeneous equation has general solution

$$y_h = c_1 y_1 + c_2 y_2, \text{ where } c_1, c_2 \in \mathbb{R}.$$

To construct a particular solution to the non-homogeneous equation, we *vary the parameters* of y_h: that is, we consider an expression of the form

$$y = uy_1 + vy_2.$$

We wish to determine conditions on the functions $u(x)$ and $v(x)$ such that the above expression is a solution to the non-homogeneous equation.

Observe that its first derivative is given by

$$y' = u'y_1 + uy_1' + v'y_2 + vy_2'.$$

We now make a certain assumption: in order to simplify our expression for y', let us suppose that we can choose u and v such that $u'y_1 + v'y_2 = 0$.* Then, we may now compute the second derivative:

$$y' = uy_1' + vy_2' \quad \Longrightarrow \quad y'' = u'y_1' + uy_1'' + v'y_2' + vy_2''.$$

We can now substitute our expressions for y, y', and y'' into the non-homogeneous differential equation:

$$\left[u'y_1' + uy_1'' + v'y_2' + vy_2''\right] + p\left[uy_1' + vy_2'\right] + q\left[uy_1 + vy_2\right] = f.$$

Rearranging terms, we have

$$u\left[y_1'' + py_1' + qy_1\right] + v\left[y_2'' + py_2' + qy_2\right] + u'y_1' + v'y_2' = f.$$

Since y_1 and y_2 are solutions to the *homogeneous* differential equation, the expressions in square brackets must be equal to zero. Our equation thus reduces to

$$u'y_1' + v'y_2' = f.$$

We hence arrive at two conditions that the functions u and v must satisfy:

$$\left\{u'y_1 + v'y_2 = 0, \quad u'y_1' + v'y_2' = f\right\}.$$

* There is no good reason for us to make this assumption, other than to simplify our expression for y'. We will soon see, however, that this works in our favor.

Solving this system of equations determines the conditions that u and v must satisfy. First, we can equate our two expressions for v'

$$v' = -\frac{u'y_1}{y_2} = \frac{f - u'y_1'}{y_2'}.$$

This equality allows us to obtain an expression for u':

$$u' = -\frac{y_2 f}{y_1 y_2' - y_1' y_2} \quad \Longrightarrow \quad u = -\int \frac{y_2 f}{y_1 y_2' - y_1' y_2}\, dx.$$

Similarly, we can obtain an expression for v':

$$v' = \frac{y_1 f}{y_1 y_2' - y_1' y_2} \quad \Longrightarrow \quad v = \int \frac{y_1 f}{y_1 y_2' - y_1' y_2}\, dx.$$

These are precisely the formulae for $u(x)$ and $v(x)$ that yield a particular solution to the non-homogeneous equation.

Table of Laplace Transforms

$f(t) = \mathcal{L}^{-1}[F(s)]$	$F(s) = \mathcal{L}[f(t)]$
1	$\frac{1}{s}$
e^{at}	$\frac{1}{s-a}$
t^n, for $n \in \mathbb{N}$	$\frac{n!}{s^{n+1}}$
$\sqrt{t}$	$\frac{\sqrt{\pi}}{2s^{3/2}}$
$\cos at$	$\frac{s}{s^2+a^2}$
$\sin at$	$\frac{a}{s^2+a^2}$
$u(t-c)$	$\frac{e^{-cs}}{s}$
$\delta(t-c)$	e^{-cs}

Properties of the Laplace Transform

We present proofs of some key properties of the Laplace Transform.

Theorem (Differentiation Property)**.** *Let $F(s)$ be the Laplace transform of $f(t)$. Then,*

$$\mathcal{L}[tf(t)] = -F'(s).$$

Proof. Let $F(s) = \mathcal{L}[f(t)]$. Differentiating this expression with respect to s, we have

$$\begin{aligned} F'(s) &= \frac{d}{ds}\lim_{h\to\infty}\int_0^h e^{-st}f(t)\,dt \\ &= \lim_{h\to\infty}\int_0^h (-t)\,e^{-st}f(t)\,dt \\ &= -\lim_{h\to\infty}\int_0^h e^{-st}\left[tf(t)\right]dt \\ &= -\mathcal{L}[tf(t)]. \end{aligned}$$

□

Theorem (The First Shifting Theorem)**.** *Let $F(s)$ be the Laplace transform of $f(t)$. Then,*

$$\mathcal{L}\left[e^{at}f(t)\right] = F(s-a).$$

Proof. We directly apply the definition of the Laplace transform:

$$\begin{aligned} \mathcal{L}\left[e^{at}f(t)\right] &= \lim_{h\to\infty}\int_0^h e^{at}e^{-st}f(t)\,dt \\ &= \lim_{h\to\infty}\int_0^h e^{-(s-a)t}f(t)\,dt \\ &= F(s-a). \end{aligned}$$

□

Theorem (Laplace Transform of Derivatives)**.** *Suppose that $f(t)$ is continuous with a well-defined Laplace transform, and that $f'(t)$ is piecewise continuous. Then,*

$$\mathcal{L}[f'(t)] = s\mathcal{L}[f(t)] - f(0).$$

Proof. We directly apply the definition of the Laplace transform:

$$\mathcal{L}[f'(t)] = \lim_{h\to\infty}\int_0^h e^{-st}f'(t)\,dt.$$

Integrating by parts, let $u = e^{-st}$ and $dv = f'(t)\,dt$. Then, $du = -se^{-st}dt$, $v = f(t)$, and

$$\begin{aligned} \mathcal{L}[f'(t)] &= \lim_{h\to\infty} e^{-st}f(t)\Big|_0^h - \int_0^h \left(-se^{-st}\right)f(t)\,dt \\ &= -f(0) + s\lim_{h\to\infty}\int_0^h e^{-st}f(t)\,dt \\ &= -f(0) + s\mathcal{L}[f(t)]. \end{aligned}$$

□

Theorem (The Second Shifting Theorem)**.** *Let $F(s)$ be the Laplace transform of $f(t)$. Then,*

$$\mathcal{L}[f(t-c)\,u(t-c)] = e^{-sc}F(s).$$

Proof. By the definition of the Laplace transform,

$$\begin{aligned}\mathcal{L}[f(t-c)\,u(t-c)] &= \lim_{h\to\infty}\int_0^h e^{-st}f(t-c)\,u(t-c)\;dt\\ &= \int_0^c e^{-st}f(t-c)\,u(t-c)\;dt\\ &\quad + \lim_{h\to\infty}\int_c^h e^{-st}f(t-c)\,u(t-c)\;dt.\end{aligned}$$

Since $u(t-c) = 0$ when $t < c$, the first integral disappears. To compute the second integral, note that $u(t-c) = 1$ when $t > c$. We proceed with a substitution: let $\tau = t-c$, with $d\tau = dt$. Then,

$$\begin{aligned}\mathcal{L}[f(t-c)\,u(t-c)] &= \lim_{h\to\infty}\int_{c-c}^{h-c} e^{-s(\tau+c)}f(\tau)\;d\tau\\ &= \lim_{h\to\infty}\int_0^{h-c} e^{-s\tau}e^{-sc}f(\tau)\;d\tau\\ &= e^{-sc}\cdot\lim_{h\to\infty}\int_0^{h-c} e^{-s\tau}f(\tau)\;d\tau\\ &= e^{-sc}F(s).\end{aligned}$$

□

Index

Bernoulli equation, 25

Dirac delta function, 49
direction fields, 17

equilibrium solutions, 18

harmonic oscillator
  damped motion, 38
  resonance, 40
  solution, 37
heat equation, 55
higher-order equations
  complex roots, 28
  distinct roots, 28
  mixed roots, 29
  repeated roots, 28

integrating factors, 24

Laplace transform
  definition, 41
  Differentiation Property, 42
  First Shifting Theorem, 43
  initial value problems, 43
  inverse, 42
  Second Shifting Theorem, 46

partial differential equations
  separation of variables, 53
  solutions, 51
  superposition principle, 52
problems from physics
  falling object with drag, 26
  Newton's Law of Cooling, 13
  radioactive decay, 12

separation of variables
  y/x-substitution, 16
  linear change of variable, 15

undetermined coefficients
  exponentials, 32
  modifications, 34
  polynomials, 32
  products, 35
  trigonometric functions, 33
unit step function, 47

variation of parameters, 35
Verhulst model
  solution, 19
  with harvesting, 20

You've made it to the end!



Illustration of a vintage nobleman,
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